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Community Disaster Preparedness App

Bsc (Hons) Computer Science and Software Engineering Final
Report

Department of Computer Science and Technology
University of Bedfordshire

Supervisor: Pawan KC

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ABSTRACT

Nepal frequently suffers untimely natural disasters such as earthquakes, floods, and landslide which leads to several loss of life and property. Many people especially, in remote areas, they struggle to get timely notifications and access essential emergency resources due to poor internet connectivity and limited preparedness guidelines. This research focuses on creating a disaster preparedness mobile application that uses real-time alert APIs, geolocation services, and offline access to keep users informed and safe during emergencies.

The app will focus on data from reliable disaster alert APIs, geolocation technologies to improve the efficiency of the proposed disaster preparedness application. Disaster alert APIs from DHM Nepal, GDACS, and OpenWeather will be used to send accurate, real-time notifications. Navigation and geolocation services such as the Google Maps API and OpenStreetMap will help users locate shelters, emergency contacts and preparedness guide at the time of the disaster. The app will work even with no internet connectivity by storing important information offline. Users can report incidents, which will be verified by the admin to avoid false information.

This mobile application will have a user-friendly interface, personalized alert settings, checklists, and shelter locator in map. The app aims to reduce disaster risks, improve community response, and support both rural and urban areas with reliable and fast information during critical times.

Keywords: Disaster alert systems, Risk reduction, Geolocation in disaster management, Offline emergency access, Emergency response, Mobile technology etc.

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Bsc. CS & SE

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List of Abbreviations

Abbreviation	Full Form	Description
API	Application Programming Interface	Enables communication between software components
AES	Advanced Encryption Standard	Symmetric encryption algorithm for data security
DHM	Department of Hydrology and Meteorology (Nepal)	Provides weather/disaster data for Nepal
FEMA	Federal Emergency Management Agency (USA)	Disaster management guidelines provider
GDACS	Global Disaster Alert and Coordination System	Real-time global disaster alert platform
GPS	Global Positioning System	Satellite-based navigation system
GIS	Geographic Information System	Mapping and spatial analysis technology
OAuth	Open Authorization	Secure authentication framework
OTP	One Time Password	Dynamic password for secure access
SQLite	Structured Query Language Lite	Lightweight database engine
UI/UX	User Interface/User Experience	Design of user interactions

Table 1: List of Abbreviation

Chapter 1: Introduction

1.1 Background

Natural disasters such as earthquakes, floods, and landslides causes significant threats to human lives, including infrastructure, economies and environmental surroundings. No proper knowledge and delayed response often lead to harsh consequences. With the advancement of technology, mobile applications have become a crucial tool in disaster management by providing real-time alerts, emergency resources, and communication platforms. This Disaster Preparedness Mobile Application is designed to fill the gap between disaster warnings and user readiness by delivering timely notifications, emergency checklists, and offline functionality to help individuals and communities respond effectively.

Recent research have highlighted the significance and efficacy of such applications. For example, a 2024 scoping assessment investigated the use of mobile apps for mental health responses to natural disasters, emphasizing their potential for providing mental health support prior to, during, and after disasters (Ezeonu, 2024). Another extensive study examined 45 disaster support mobile apps and over 28,000 user ratings, highlighting essential features and user feedback to help enhance future app development (Muahamad, Anuradha, Mojtaba, & John, 2024). Furthermore, research on the acceptance of mobile applications for disaster preparedness simulation demonstrated their potential to build decision intelligence in disaster management (James , et al., 2023).

1.2 Problem Statement

In many communities, particularly in urban areas, people face significant challenges in receiving timely alerts and accessing critical emergency information. This is frequently due to lack of centralized information systems and insufficient connectivity, which can lead to delayed responses and greater danger, huge loss of life and property along with

the surroundings. People are not aware of such effective warning mechanisms, preparedness guidelines and tools. Furthermore, the lack of a unified command structure and insufficient information exchange among emergency resources may worsen the issue. These challenges highlights the importance of a strong communication infrastructure and coordinated emergency management strategies to enhance more preparedness among communities of both rural and urban areas (Seyed, Ardashir, Davoud, & Mahsa, 2021).

1.3 Proposed Solution

To address the issues of delayed notifications and a lack of centralized emergency information, a Comprehensive Community Disaster Preparedness Mobile Application is proposed. This app will serve as a centralized information hub, collecting real-time data from government agencies, meteorological departments, and emergency services to guarantee the users receive accurate and fast alerts. This software will use geolocation technology to provide location specific notifications, ensuring that users are alerted about specific disasters without being notified by unnecessary warnings. In addition, to address internet connectivity issues during emergencies, this application will allow users to access critical resources such as evacuation routes, emergency contacts, and first aid recommendations for offline usage.

By enabling users to report local incidents occurred while, after the disaster in real-time, the app will also promote community engagement and community-driven approach to disaster response. Additionally, it will collaborate with emergency services to offer direct lines of communication for disaster alerts and management assistance. The Emergency Management Connect App, a similar application has demonstrated how well digital technology can enhance communication between people and emergency services by passing critical resources (Pulsara, 2024). The proposed mobile application aims to decrease the dangers in both rural and urban areas, improving response times, and raise awareness about disaster by including these components.

1.4 Aim and Objectives

Aim:

- The major aim of this research is to investigate how modern mobile technologies can enhance community disaster preparedness by delivering a reliable, efficient, and user-friendly disaster alert and response system. It seeks to promote user safety and awareness by creating an accessible, simple yet effective mobile application that supports fast information delivery and informed decision making during emergencies.

Objectives:

- Develop a mobile application with real-time alerts and notifications
- Make essential features available offline for reliability
- Provide emergency preparedness
- Secure data storage and authentication

1.5 Feasibility Study

The Community Disaster Preparedness Mobile Application feasibility study assesses the project's technical, economic, operational, and legal viability to assure its effective implementation.

Technically, this application is developed for Android, which is mostly used in Nepal. The app uses Firebase to manage user accounts, store incidents reports and secure the data. It also uses disaster warning APIs (GDACS and OpenWeather) to deliver real-time data. This program includes a flutter_map_tile_caching for offline use, allowing users to view shelters stored in the map and access them when there is no internet connection.

Economically, the project is cost-effective because it mostly uses open-source technology and publicly available disaster API, which lowers development expenses. If necessary,

revenue tactics such as premium features, NGO partnerships, or government collaborations can ensure the app's long-term viability.

Operationally, the application is simple, easy to use, and designed for people. Its clean interface, clear navigation and offline access make it helpful during emergencies when users need quick and reliable information. These features ensures that the app can be used effectively in a stressful situation.

Legally, the app follows basic data protection and privacy requirements. This app follows principles similar to GDPR, ensuring that user data is safe and reponsible. Firebase provides secure authentication, encrypted data storage and strong access to protect user information.

1.6 Intellectual Challenges

To maintain overall safety, security, and usability, the Community Disaster Preparedness Application has encountered several technical issues that has been addressed. Because the application handles real-time notification, offline mode and data accuracy. The following sections highlight the major system-related issues discovered during the development process.

1.6.1 Real-Time Data Integration and Synchronization

One of the challenges I faced was using multiple disaster data sources for real-time alerts. My first aim was to combine three validated APIs (GDACS, DHM Nepal and Open Weather) to provide reliable and diveres disaster information. However, I could only use two GDACS and Open Weather in the final system because when I contacted DHM Nepal, they advised me that access to their official disaster API needed an MoU between the institution which was not possible due to this I was unable to implement their data.

1.6.2 Offline Functionality and Map Stability

Essential information such as shelter locations, emergency contacts, and preparedness checklists needed to be accessible when there is no internet connection, which is common in disaster situations. To ensure that offline data remains updated and synchronized without hampering the stored information required a well designed local storage system.

Similarly, a stable map support was challenging. During the development of this application I tested multiple map libraries like OpenStreetMap, Mapbox, Google Maps, etc. Most of these maps were crashing, rendering issues, or the app failed. These problems often occurred during offline mode or while loading the app 2-3 times. The final working solution was achieved by using `flutter_map_tile_caching` ((Luka), 2025), which supported stable offline tile storage and also to maintain performance on low end Android devices.

1.6.3 Cross-Platform Limitaion (Testing and Building for iOS)

Another major challenge I faced was the limitation in testing and building the application for iOS. Although Flutter lets the program to operate on both Android and iOS version requires a macOS system, Xcode, and an Apple Developer account. Because I did not have access to a macOS device, I was unable to develop an iOS version using Android Studio, which made testing difficult.

As a result, I was only able to build, test, and optimize the program on Android devices. This restricted cross-platform support and prevented me from testing how the software performed on iPhones. Despite the app's technical ability to function on iOS, the absence of essential hardware and environment made it difficult to fully create an iOS version within the project timeline.

1.6.4 User Accessibility and Interface Challenges

Another important requirement was to create a user interface that is functional even under stressful situations. The software was made for diverse users, including elderly, those with limited access to modern world. This necessitates simple navigation, fonts, clear design, and accurate alerts. Maintaining accessibility and usability required UI/UX testing, user feedback collection, and interface modification to avoid confusion during emergency.

1.7 Thesis Report Format

Chapter	Contents
Chapter 1- Introduction	- Background - Problem Statement - Proposed Solution - Aim and Objectives - Feasibility Study - Intellectual Challenges - Thesis Format
Chapter 2 – Literature Review	- Overview of Disaster Preparedness Technologies - Existing Disaster Preparedness Mobile Applications - Real-Time Alert Systems and APIs - Geolocation and Emergency Response Technologies

	- Offline Functionality in Mobile Apps - Data Security and Privacy - Summary of Literature Findings
Chapter 3 – Methodology	- Research Approach and Design - Data Analysis - System Development Tools and Technologies - Disaster Alert API Integration - Offline Data Storage Implementation - Security Implementation (Oauth 2.0, AES Encryption) - UI/UX Design Considerations - Testing and Validation Approach
Chapter 4 – System Design and Implementation	- System Architecture Overview - Use Case Diagrams and System Flowcharts - Database Design and Structure - API Integration for Real-Time Alerts - Geolocation and Shelter Mapping System - Offline Mode Development - Secure Authentication and User Data Management
Chapter 5 – Results and Discussion	- System Testing and Performance Analysis - User Feedback and Usability Testing

	- Evaluation of Disaster Alert Accuracy - Comparison with Existing Systems - Challenges Faced During Development - Recommendations for Improvement
Chapter 6 – Conclusion and Future Work	- Summary of Findings - Contributions of the Research - Limitations of the Study - Future Enhancements and Recommendations
References	- List of all the academic papers, books, and online sources cited in Harvard Referencing Style
Appendix	- Survey Questions and User Feedback Forms - Additional Diagrams and Screenshots of the App - Code Snippets

Table 2: Thesis Format

Chapter 2: Project Plan

2.1 Gantt Chart

A Gantt chart is a project management tool that helps with the planning and scheduling of projects of all sizes; it is especially good for visualizing projects. A Gantt chart is a

graphical representation of activity versus time that helps project managers track progress (Management, 2025).

2.1.1 Semester 1 Gantt Chart

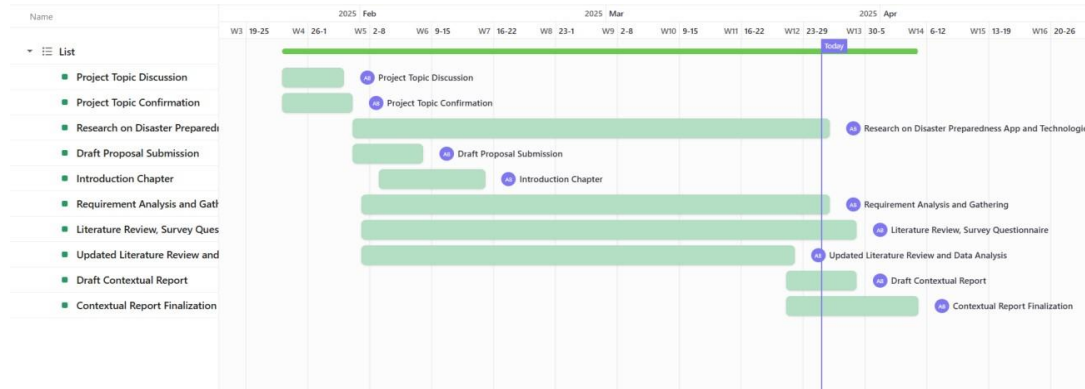


Figure 1: Gantt Chart of Semester-1

The project schedule for semester 1, which runs from January to March 2025, is shown in the Gantt chart above. The whole project task is shown by the green bar.

2.1.2 Semester 2 Gantt Chart

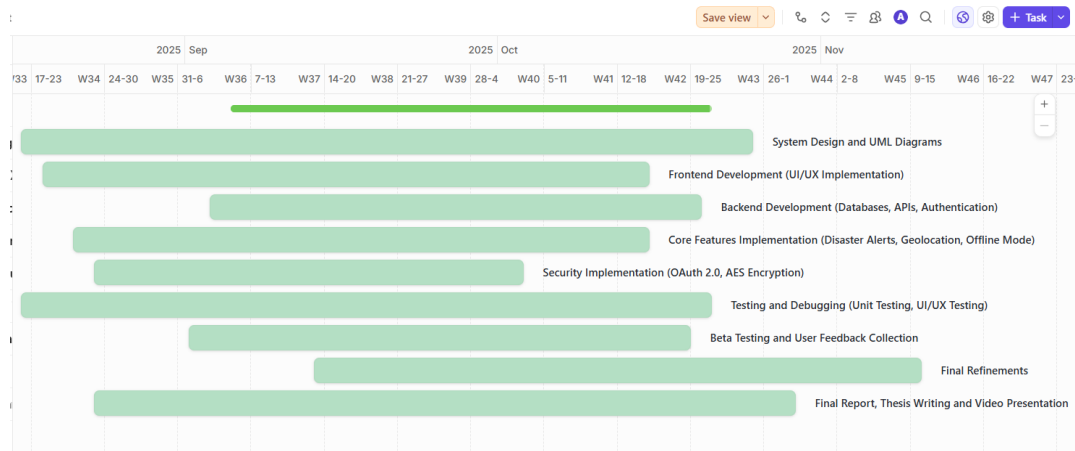


Figure 2: Gantt Chart of Semester-2

This above figure was the project timeline from September to November 2025 detailing every work done. This helped me in my project's completion because I had a clear roadmap.

2.2 Risk Analysis

Risk analysis is the process of finding, assessing, and comprehending probable events that could harm a business, project, or operation. It entails calculating the chance of those risks occurring, the potential consequences, and the total threat level they represent (Stream, 2025).

Risk Analysis	Impact	Likelihood	Mitigate risks
API reliability issues affecting real-time alerts.	High	Likely	Implement backup alert source for redundancy. (adding alternative source if primary alert system fails)
High data dependency on external source.	High	Likely	Minimize external dependencies by storing local data.
User engagement challenges.	High	Likely	Develop interactive features to improve user engagement.
Cyber security threats and data privacy concerns.	High	Likely	Implement encryption, authentication protocols, and secure database management
Accessibility issue in remote areas.	High	Highly Likely	Ensure offline functionality and low-data mode.

Table 3: Risk Analysis

2.3 Workflow Diagram

Workflow diagrams (or flowcharts) are visual infographics that depict the sequential flow of processes or tasks in a process. They serve as road maps for getting tasks done (ATLASSIAN, 2025).

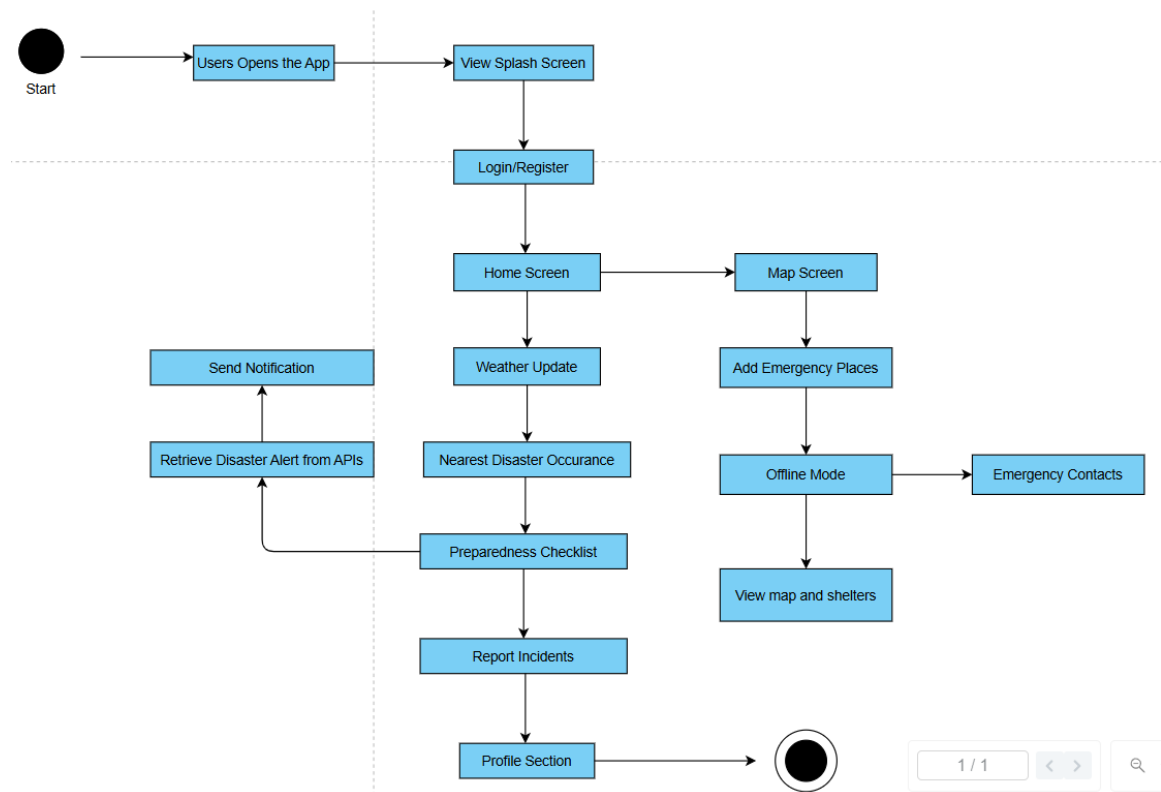


Figure 3: User workflow diagram

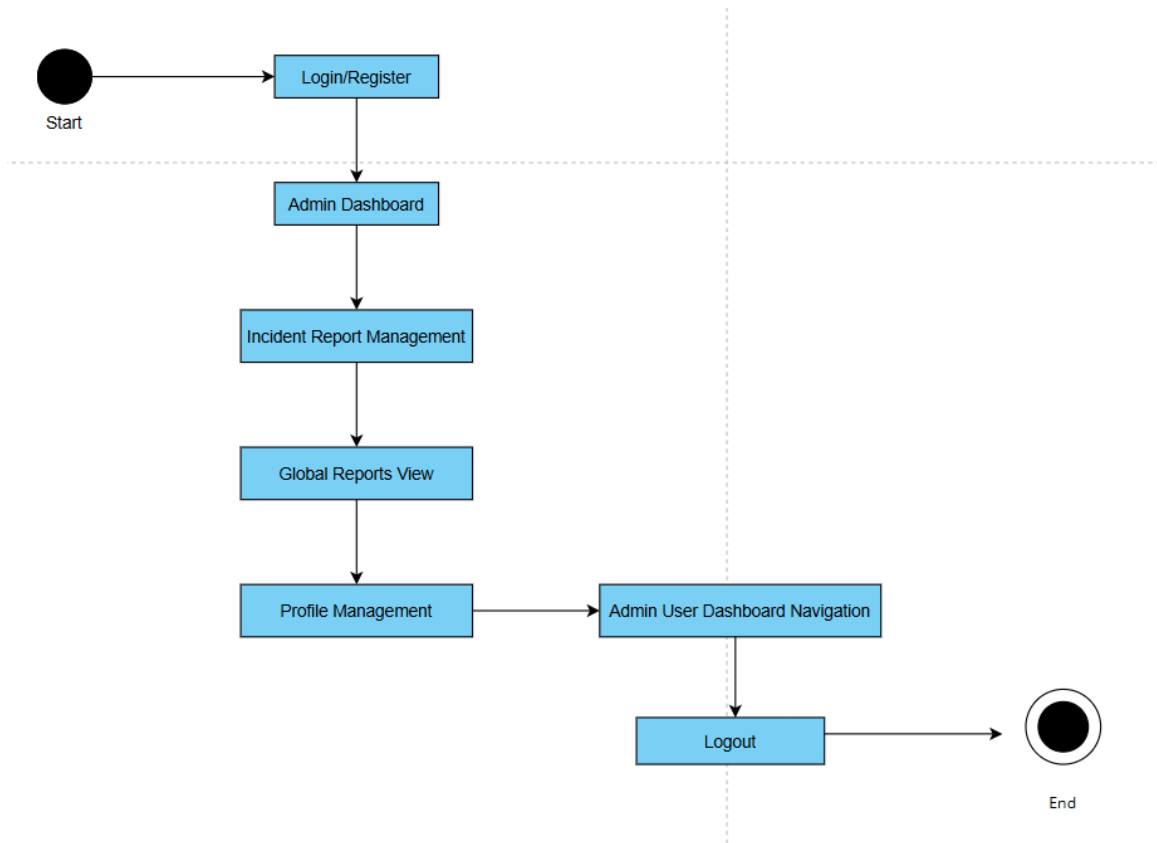


Figure 4: Admin Workflow Diagram

Chapter 3: Literature Review (LR)

3.1 Overview/Background

The environment, infrastructure, and human lives are all seriously at risk from natural disasters including landslides, floods, and earthquakes. The impact of these calamities is frequently exacerbated by insufficient emergency planning, a lack of early alerts, and a lack of resource coordination. Because they provide offline accessibility, emergency checklists, geolocation services, and real-time notifications, mobile applications have become essential in disaster management as a result of technological improvements.

Applications for disaster preparedness have been created by a number of public and private entities to increase community resilience. These applications incorporate

geolocation tracking, emergency communication capabilities, disaster alarm APIs, and educational materials. However, issues including poor user engagement, security problems, delayed notifications, and lack of offline capabilities continue to plague current solutions. The goal of this project is to improve disaster readiness by developing a mobile application that combines secure user data management, emergency resources, and real-time warnings.

3.2 Review of Existing Project

3.2.1 Bipad Portal

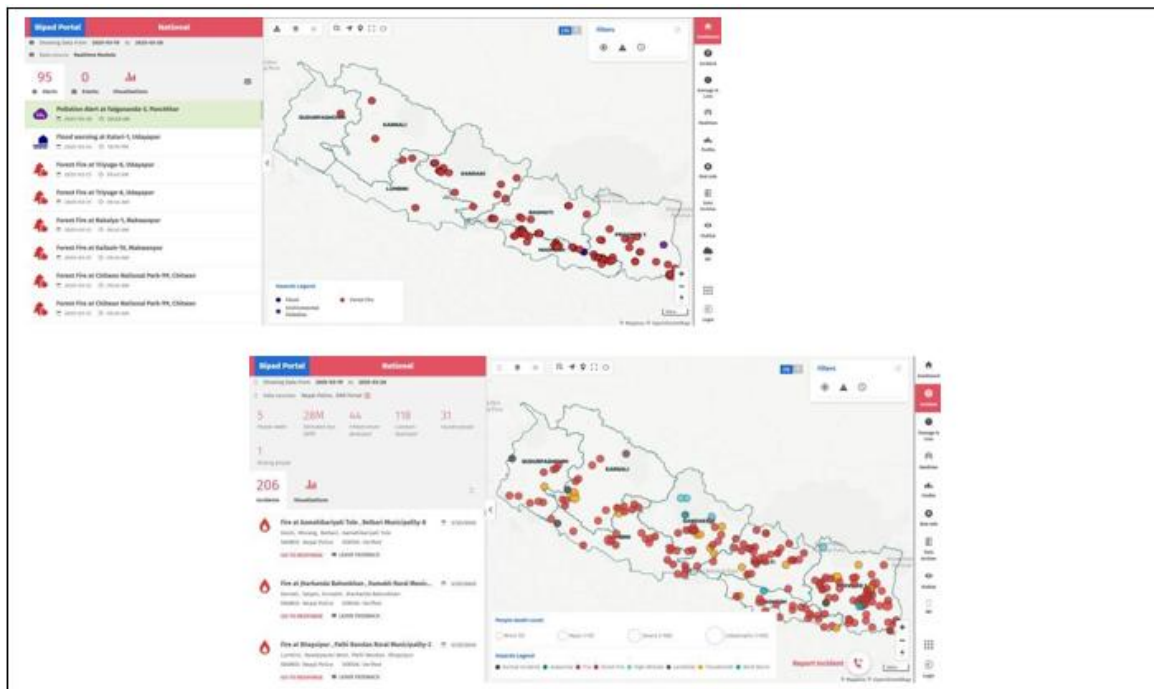


Figure 5: Bipad Portal

Bipad Portal is a government web based disaster risk management system developed by Government of Nepal to provide real-time disaster alerts, risk mapping, and access to emergency policies. This portal lacks offline functionality and does not allow users to customized alerts based on specific locations or disaster preferences.

3.2.2 Pahirol Alert

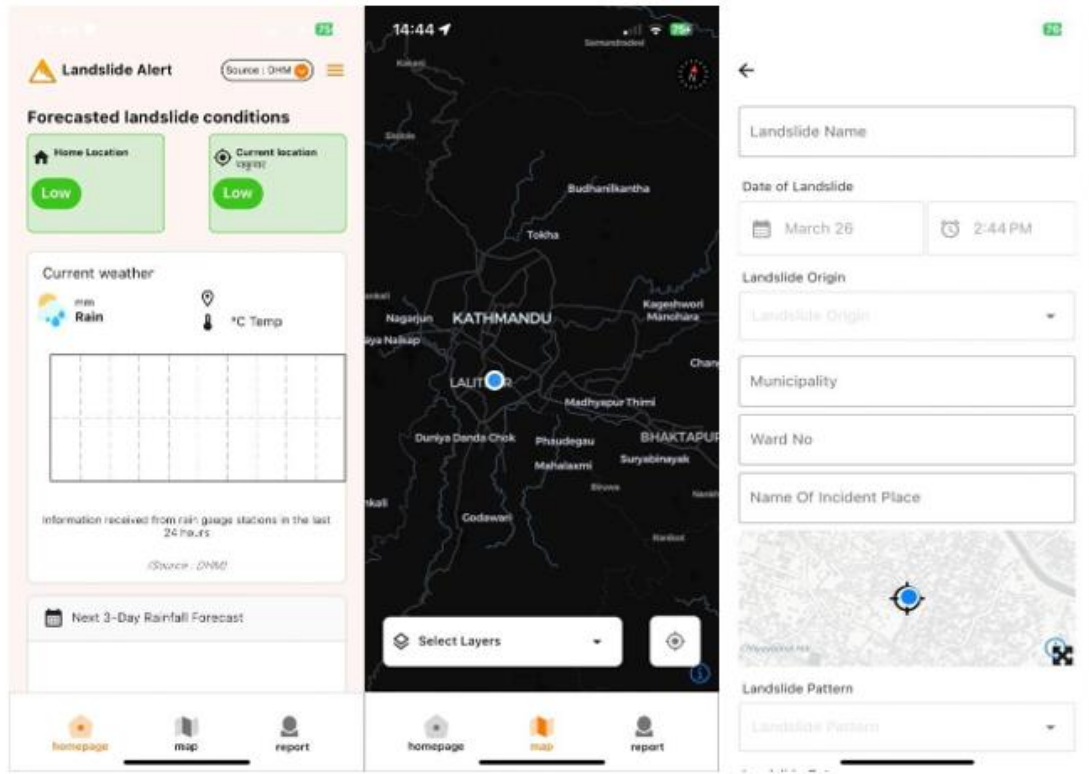


Figure 6: Pahirol Alert

Pahiro Alert is a mobile application developed by DHM Nepal to provide early warnings for landslide and floods. This application utilizes real-time weather tracking and scientific models to predict potential landslide and flood prone areas. This has limited options because only 2 disaster types are there and lacks a shelter mapping features.

3.2.3 First Aid – IFRC

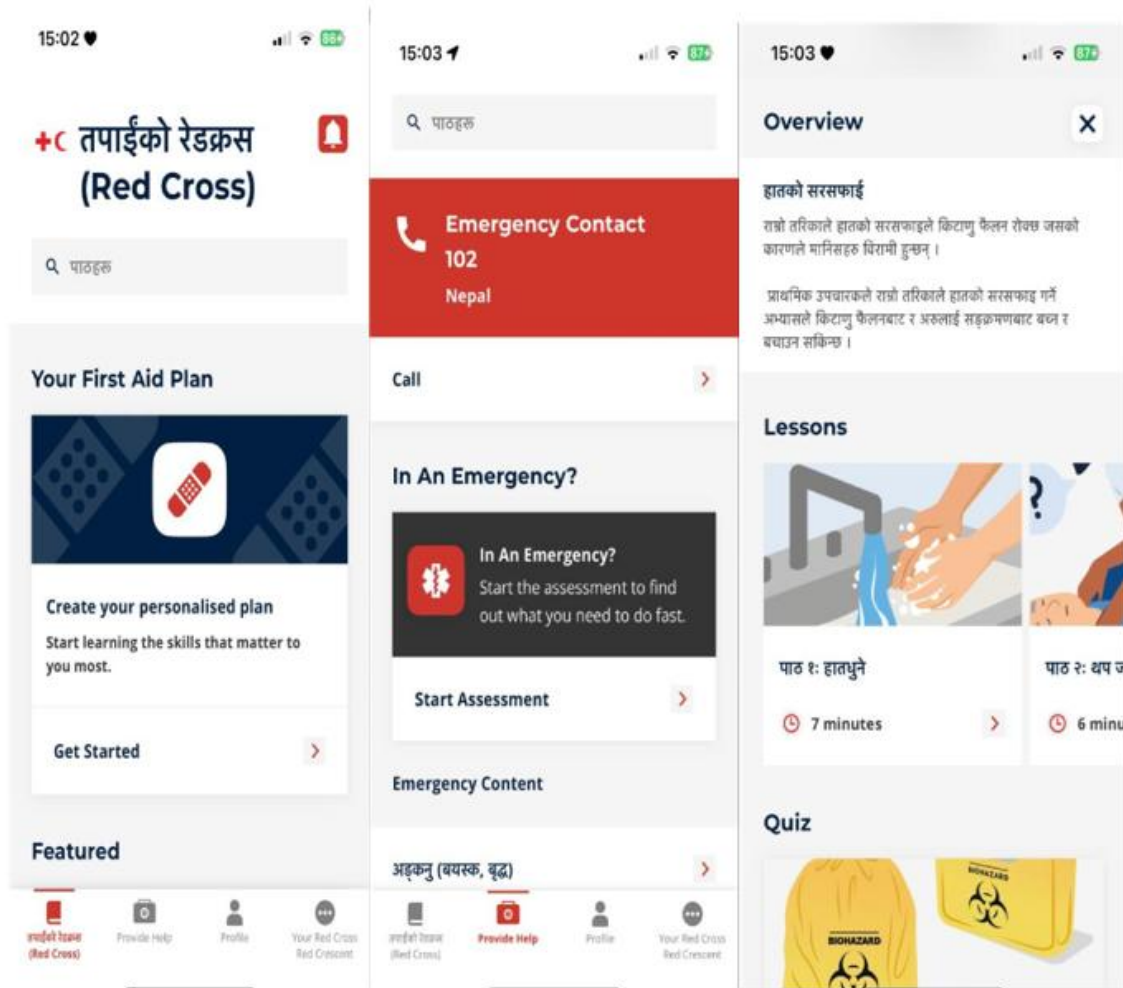


Figure 7: First Aid – IFRC

The International Federation of Red Cross and Red Crescent Societies (IFRC) is a mobile based application developed by the First Aid that provides life-saving first aid instructions for a variety of emergency circumstances. The software features interactive step-by-step guidelines, videos, and quizzes that teach users how to deal with medical situations. However, it lacks real time disaster alerts functions, shelter mapping, and emergency assistance.

3.2.4 Bachau App

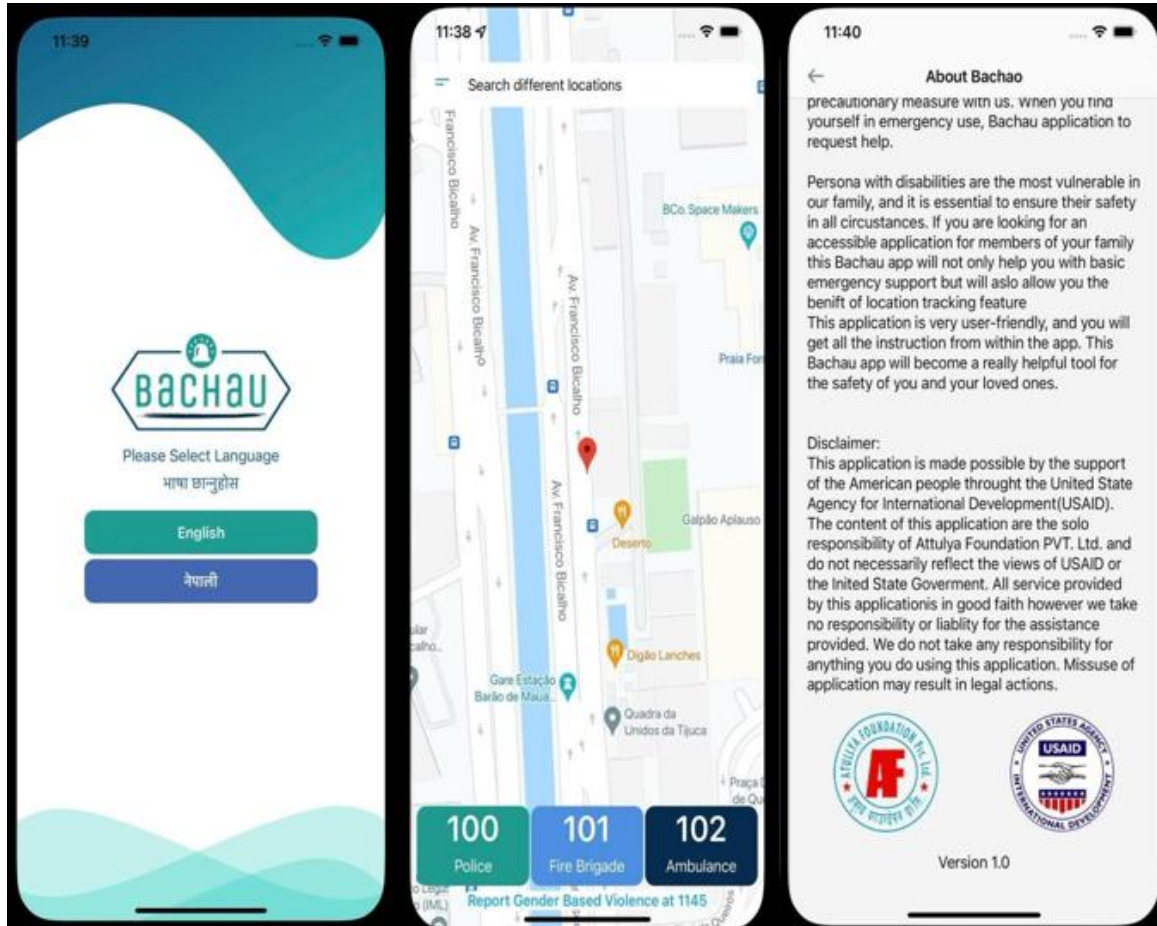


Figure 8: Bachau App

Bachau App is a mobile based application that connects users to emergency services such as police, fire brigades, ambulances and hospitals. This app has one touch emergency assistance, emergency contact integration, and location sharing as its key features where it lacks real-time alerts, offline functionality and disaster preparedness guides.

3.3 Key Terms Used

1. Technical Terms

1.1 API (Application Programming Interface)

- An application programming interface (API) is code that enables two software programs to communicate (Alexander S., Ben, & Tom, 2025).

1.2 Geolocation Technologies

- Technologies like GPS, or mobile network-based systems used to identify the real world geographic location of a device.

1.3 GDACS (Global Disaster Alert and Coordination System)

- A platform that provides real-time alerts about natural disasters worldwide (GDACS, 2025).

1.4 Oauth 2.0 (Open Authorization)

- It is a safety protocol that lets apps access your data from another service without needing your password. (E.g., Google or Facebook) (OAuth 2. , 2023)

1.5 AES (Advanced Encryption Standard)

- Encryption of data so hackers cant read it. (Eg., WhatsApp private messages, AES encrypts it so only the receiver can read it) (Rahul , Corinne, & Michael, 2025)

1.6 Firebase

- A Google-backend platform for developing mobile and web applications, offering cloud storage, authentication, and real-time databases (Firebase, 2023).

1.7 SQLite

- A lightweight, file-based database system used for local (offline) data storage in apps (SQLite, 2025).

1.8 Flutter

- A open source framework for building cross-platform mobile applications (Android and iOS) (Flutter, 2025).

1.9 Use Case Diagram

- A visual representation of how users interact with a system, showing different functions of the system (techtargt, 2020).

1.10 UI/UX (User Interface/User Experience)

- UI refers to the visual design of an app, while UX focuses on usability and user interaction.

1.11 Offline Caching

- Storing data locally on device so that app functionalities can be used without an internet connection (Mold & Ana, 2024).

1.12 Data Synchronization

- The process of ensuring that data across multiple devices or servers stays updated and consistent.

2. Disaster Management Terms:

2.1 Disaster Risk Reduction (DRR)

- Preparing for disasters before they happen (e.g., drills of natural disaster)

2.2 Situational Awareness

- Knowing what is happening in a disaster (e.g., app showing flood zones)

2.3 Evacuation Route

- A safe path to escape danger (e.g., safe zones, emergency exit signs)

2.4 Community Resilience

- How fast a city or place recovers after a disaster (e.g., rebuilding after a disaster has occurred)

2.5 Shelter Mapping

- App showing safe places nearby (like schools turned into storm shelters)

2.6 Incident Reporting

- Users sending disaster photos/ locations to the app (like reporting a landslide)

3.4 Literature Review

3.4.1 The Synergy of Emergency Alerts and Social media: An Evaluation with the Emergency Alert and Social Media Engagement Test (Digvijay, Ramesh, & Rishab, 2024)

Using the “Emergency Alert and Social Media Engagement Test (EASE Test)”, this study investigates the relationship between traditional emergency alerts and social media. The study focuses on how social media improves emergency communication by facilitating real-time information sharing, increasing situational awareness, and encouraging public participation. The key findings show that participants who actively used social media during simulated emergencies responded more effectively to notifications because they were better informed and took prompt action. However, concerns such as disinformation and data privacy were identified. The study recommends an integrated approach that combines the benefits of traditional alerts and social media to increase public safety and disaster resilience.

Inspired by the research findings, my app will combine real-time emergency alerts with social media interaction features to improve user responsiveness and situational awareness. It will send out clear, timely alerts while also allowing people to submit a form which will contain photos to prevent false information during a crisis.

3.4.2 A Systematic Review of Disaster Management Systems: Approaches, Challenges, and Future Directions (Saad, Imran, Wasi, & Isabel , 2023)

Using “A Systematic Review of Disaster Management Systems”, this literature review investigates the use of sophisticated technologies such as remote sensing, GIS, IoT, and machine learning in disaster management systems for earthquakes, wildfires, and floods. Studies show that real-time monitoring and predictive modeling are effective for early warning systems, while remote sensing provides crucial data for situational awareness.

Limited datasets, communication obstacles, and the need for community engagement are some of the challenges. Researchers emphasize the significance of integrated systems, which incorporate numerous technologies to improve catastrophe preparedness, response, and resilience. Future directions include developing standardized frameworks, improving data accuracy, and encouraging community participation, all of which will provide significant insights for creating more effective disaster management systems.

Based on the research findings, my app will use Google Maps and OpenStreetMap to offer real-time updates on natural disaster. These mapping tools will help depict disaster-prone locations, evacuation routes, and safe zones, assisting with decision-making and resource allocation. To encourage community engagement, the app will contain capabilities for crowdsourced data collection and community alerts, allowing users to share real-time updates and participate during emergencies. Using these mapping capabilities and community-driven features, the app intends to develop a unified platform for cross-disaster management, improving preparedness, response, and resilience in disaster-prone areas.

3.4.3 Enhancing Emergency Response Management using Emergency Description Information Technology (EDIT): A Design Science Approach (Michael & Will, 2021)

The paper explores the mobile application "Emergency Description Information Technology (EDIT)" as a solution to optimize emergency response management through improved crucial information handling during emergencies. The research shows that conventional emergency phone systems yield poor performance through wrong location reports alongside language barriers and delayed support operations. Mobile technology in EDIT automatically gathers essential data that dispatchers receive to boost response efficiency while reducing errors in data collection. The design-science methodology used for creating the prototype demonstrates how visual and automated data collection mechanisms can bring value to conventional speech communication interfaces. The research validated the necessity of building interfaces that present simple designs and

remain relaxing throughout while also providing flexible procedures to handle different emergency situations.

The app I developed as an EDIT copy uses Google Maps and OpenStreet Map to gather emergency data in real time which shows dangerous areas while giving precise location information to users. The application features an easy-to-use interface which presents emergency reporting questions in simplified steps to help users maintain calmness.

3.4.4 Designing Apps with Offline Caching for Enhanced Performance and User Experience (Mold & Ana, 2024)

The research paper “Designing Apps with Offline Caching for Enhanced Performance and User Experience” demonstrates how offline caching enhances user experience because it delivers faster app functions and maintained operational abilities during periods with weak internet connectivity. Research reveals that mobile apps with built-in offline functionality speed up loading times by approximately 80% while both reducing abandonment rates and strengthening user retention by 30%. The effective implementation of performance-enhancing tactics includes deploying Indexed DB alongside Service Workers and SQLite as well as executing data synchronization properly and prioritizing cached data according to use frequency. Data storage limits and consistency along with security issues can be solved through powerful synchronization systems, encryption algorithms and periodic audit assessments. Workbox Pouch DB and Cache API serve as popular tools for effective offline caching while the dependability of devices and networks is checked by Cypress as well as Jest testing frameworks. The system will use SQLite for offline data storage that provides local critical disaster-related information preservation including shelter locations emergency contacts and checklists.

The system enables users to access information without interruption under any connectivity situation. Service Workers improve offline functionality through their ability to store static resources which include maps and guidelines. Through this approach users

will achieve uninterrupted access to vital information which enhances disaster response efficiency and trust level.

3.4.5 Mobile Application for Disaster Safety Management (M Siva, Harshitha , Vadaparthi, & Vishnu , 2024)

The article investigates how developers built a disaster safety mobile application to assist Indian people and communities during natural disasters that include earthquakes and floods and cyclones. The application development depends on Android Studio with Firebase Realtime Database for real-time data synchronization to deliver features including location-based alerts alongside emergency steps and earthquake detection via the Mercalli sensor with emergency contact access. Research demonstrates that the app sends immediate catastrophe alerts yet enables people to report standard and official emergency warnings and offers crisis-specific disaster aid. The software integrates a Mercalli sensor built into the smartphones to measure earthquake strength based on smartphone gyroscopic sensors to enhance early warning systems.

Inspired by this research it will guide my app development to integrate immediate disaster warning systems and user-generated problem reporting aimed at strengthening public readiness and emergency response. The application will use Firebase Realtime Database to ensure data synchronization which provides the most recent information about disasters and evacuation paths and emergency contacts. Users will find disaster-specific emergency steps and direct contacts to emergency services such as police and fire department in the application. The application will focus on creating an accessible platform through a simple interface design which enables users to retrieve vital information even when disconnected from the network. Similarly, the app assembles its features to enable users to serve as active protectors of disaster resilience so they can develop safety-oriented cultural practices.

3.4.6 Mobile phone technologies for disaster risk reduction (Jonathan & Emma, 2021)

This research article investigates the use of “Mobile phone technologies for disaster risk reduction (DRR)”, focusing on their potential to increase inclusivity, local relevance, and resilience in disaster-prone areas. The report emphasizes the rapid rise of mobile phone usage worldwide, particularly in developing nations, as well as its significance in enabling citizen science, real-time data collecting, and risk communications. Key findings show that mobile technologies including SMS warnings, social media platforms, and mobile apps can dramatically improve disaster preparedness, response, and recovery. However, many technologies fail because of a lack of user-centered design (UCD) and inadequate consideration of local contexts, user demands, and sustainability. The study provides a set of scalable, hazard-agnostic recommendations for exploiting mobile technologies in disaster risk reduction, with a focus on collaboration with local stakeholders, incentives, and long-term sustainability.

Inspired by this research, my app will use mobile phone technology to improve catastrophe preparedness and response. It will collect real-time data from smartphone sensors and send location-based notifications for disasters such as floods, earthquakes, and cyclones. The app will use user-centered design (UCD), which ensures that it is adapted to local contexts, languages, and literacy levels. In addition, the app will use social media integration (for example, WhatsApp, Facebook) to disseminate warnings and updates to a larger audience. The app's focus on sustainability and local relevance intends to empower users to actively participate in disaster risk reduction, transitioning them from passive recipients to active agents of resilience.

3.4.7 Exploring the Intelligent Emergency Management Mode of Rural Natural Disasters in the Era of Digital Technology (Jimei, Hanping, & Hanqing, 2024)

The research paper “Exploring the Intelligent Emergency Management Mode of Rural Natural Disasters in the Era of Digital Technology”, investigates the creation of an

intelligent emergency management mode for rural natural disasters in China by utilizing digital technologies such as artificial intelligence (AI), big data, the Internet of Things (IoT), and cloud computing. It emphasizes the issues that rural communities confront, such as poor infrastructure, inefficient information sharing, and ineffective disaster response systems. The suggested model combines modern technology to improve disaster monitoring, early warning, resource allocation, and rescue operations. Key components include an integrated "sky-air-ground" intelligent perception system, a machine learning-based emergency supply scheduling system, and a multi-dimensional resource integration strategy for rescue teams. The concept intends to increase emergency management efficiency, reduce disaster losses, and promote long-term rural development. However, problems such as technology gaps, a lack of standardization, and inadequate infrastructure in rural areas impede implementation.

Inspired by research on intelligent emergency management for rural natural disasters, my app will use readily available digital technology to improve disaster preparedness and response in rural areas. To monitor disaster-prone areas, the app will leverage real-time data collecting via user-reported inputs, GPS, and basic sensor data (where accessible). It will have a simple, user-friendly interface for rural inhabitants to report emergencies, receive early alerts, and get evacuation instructions. To increase cooperation, the app will enable communication between local authorities, rescue personnel, and communities via a single platform.

3.4.8 iĀWHINA: Towards Designing an Offline Disaster Mobile Application (Ahmed, Thelma, & Saed, 2022)

The paper describes the design stages of “iĀwhina”, a disaster response smartphone app that offers offline and online support during calamities. The program prioritizes usability, user experience, and simplicity, making it helpful in crisis scenarios where communication infrastructure may be destroyed. Offline communication via ad hoc networks, SOS warnings, location sharing, disaster information dissemination, and community participation for mutual aid are some of the key characteristics. To

accommodate people in high-stress conditions, usability studies focused on decreasing cognitive load, assuring learnability, and keeping a simple interface. The app's design process included a literature research, informal usability testing, and expert comments, emphasizing the value of user-centered design in crisis management tools. The study emphasizes the need for trustworthy, offline solutions that prioritize functionality and ease of use.

Inspired by this research paper, my application aims to provide a user-friendly, offline disaster response solution for rural areas. It will incorporate real-time location sharing, SOS warnings, and community participation for mutual aid (for example, sharing resources such as food, water, and medical supplies). The software will use ad hoc networks to communicate when standard networks are unavailable, ensuring operation in rural or disaster-stricken locations. The interface will prioritize simplicity and minimalism to lessen cognitive strain, making it simple for users to navigate during an emergency. The app will also provide disaster-related information (such as safe zones and evacuation routes) and allow users to report emergencies or request assistance. By emphasizing usability and offline functionality, my app seeks to empower rural communities to respond effectively to calamities, even without advanced AI or complex technologies.

3.4.9 Mobile Apps to Support Mental Health Response in Natural Disaster: Scoping Review (Ejemai, Medard, Janice, Petra, & Francis, 2024)

This “Mobile Apps to Support Mental Health Response in Natural Disaster: Scoping Review”, investigates the use of mobile apps for mental health responses to natural disasters, highlighting a large gap in research and application. Only five papers matched the inclusion criteria, demonstrating the minimal research into mobile apps in disaster mental health management. The reviewed apps, including Training for Life Skills (TLS), Sonoma Rises, Headspace, Psychological First Aid (PFA), and the SAMHSA Behavioral Health Disaster Response App, demonstrated promising results in addressing mental health issues such as PTSD, anxiety, and depression among disaster victims and relief

workers. These apps included psychoeducation, coping methods, symptom tracking, and crisis intervention, indicating their ability to enhance mental health outcomes. However, most research concentrated on the recovery phase of catastrophes, with limited examination of applications in preparedness and response. Barriers to adoption include accessibility challenges, privacy concerns, and cultural or linguistic constraints. The review emphasizes the importance of conducting additional research and using mobile apps into all stages of disaster management in order to improve mental health assistance.

Inspired by this research, my application will provide mental health support throughout the disaster management process, including readiness, response, and rehabilitation. It will include psychoeducational information on coping strategies and crisis support options such as hotlines and emergency numbers. To address accessibility, the software will operate offline via ad hoc networks, enabling operation even in places with low connectivity. During the preparedness phase, the app will provide mental health literacy and disaster preparedness suggestions. During the response phase, it will offer real-time assistance via SOS alerts, location sharing, and referrals to local mental health services. My app intends to provide complete mental health care to disaster-affected populations by using mobile technology's scalability and accessibility, thereby filling a major need in disaster management.

3.4.10 Acceptance of Mobile Application on Disaster Preparedness: Towards Decision Intelligence in Disaster Management (James, Arlene, Erlito, & Alexander, 2023)

This paper “Acceptance of Mobile Application on Disaster Preparedness: Towards Decision Intelligence in Disaster Management”, presents a smart disaster management architecture for the Philippines that emphasizes catastrophe preparedness using a mobile application. The app promises to improve individual and community preparedness by allowing users to develop disaster plans, emergency packs, and get real-time disaster information. The study assesses the app's adoption among disaster management personnel from non-governmental groups, indicating high levels of satisfaction with its usefulness,

convenience of use, and involvement. The suggested architecture improves decision-making and resource allocation in disaster management by combining artificial intelligence (AI), data analytics, and simulation data. Individual performance data from disaster preparedness simulations is collected by the app and evaluated to create insights for descriptive, predictive, and prescriptive analytics.

Inspired by this research, my application will focus on disaster preparedness and management by including essential aspects from the suggested architecture without relying on AI. It will contain simulationbased training modules for disasters such as earthquakes, floods, and typhoons, allowing users to practice developing evacuation plans, assembling emergency kits, and receiving real-time crisis updates. During simulations, the app will capture user performance data such as evacuation drill success rates, evaluation scores, and checklist completion rates, which will then be safely stored in the cloud for basic analysis. The app will use data analytics to provide insights to help individuals and communities prepare for disasters. It will also make individualized recommendations based on user performance and local disaster risk, employing established algorithms rather than AI.

3.4.11 iHanda: A Mobile Application for Disaster Preparedness (Ma. Corazon G., Maria, & Ace C., 2024)

This research article discusses “iHanda”, a mobile app developed to improve disaster preparedness among Manila people. The app, whose name translates as "I am prepared," aims to raise awareness and provide practical information during calamities including earthquakes, floods, fires, and typhoons. It has three primary modules: iHanda (weather forecasts, disaster preparedness infographics, vulnerability maps, and emergency hotlines), iAlam (the most recent PAGASA announcements), and iListo (maps to identify government offices, hospitals, and evacuation centers). The software also incorporates an SMS notification system, which alerts users to new announcements even when they are offline. The software, created using the Scrum approach and tested through beta testing, received favorable comments from both IT and nonIT users, with high scores for

functionality, reliability, usability, and efficiency. The study focuses on the app's potential to increase disaster preparedness in Metro Manila and offers further improvements, such as cross-platform availability and real-time notifications.

Inspired by iHanda, my application will emphasize disaster preparedness by introducing features specific to my target audience. It will feature modules for real-time weather updates, disaster preparedness infographics, and emergency contact information. My app, like iHanda, will have maps to help users find nearby hospitals, evacuation centers, and government offices. To ensure accessibility, the app will have an offline mode and SMS notifications for important updates. I will also include localized disaster information and community input tools to make the app more relevant and user-friendly.

3.4.12 Disaster Preparedness for Children and Families: A Critical Review (Eva, Briony, Kevin, & Vicotria, 2024)

This research paper “Disaster Preparedness for Children and Families: A Critical Review”, examines disaster preparedness programs for children and families, focusing on their impact in lowering risk and improving resilience to catastrophes. The analysis emphasizes that, while such programs have evolved dramatically over the last 15 years, with mostly favorable results, they face obstacles in methodological rigor, long-term effectiveness, and implementation. Key findings show that disaster preparedness education can help children improve their knowledge, attitudes, and behaviors linked to disaster risk reduction (DRR), such as comprehending protective measures and participating in home-based preparedness activities. However, the study also identifies shortcomings, such as a lack of long-term follow-up studies, insufficient information on whether these programs lower actual risk during disasters, and difficulties in scaling and maintaining program implementation.

Inspired by this research, my application will emphasize disaster preparedness for children and families, combining evidence-based components to improve both physical and mental resilience. The software will include interactive and participatory modules for

teaching children and families about disaster risks, preventative measures, and preparedness tactics. It will include critical DRR concepts like "drop, cover, and hold" for earthquakes, as well as guided activities to encourage families to discuss and practice preparedness at home. Additionally, the app will include school-based drills and community involvement tools to encourage greater participation and sustainability. Using the findings of this review, my app intends to not only increase knowledge and preparedness habits, but also to ensure that these gains convert into real-world risk reduction during disasters.

3.4.13 DISASTERPREP: Gamified Mobile Edutainment for future disaster preparedness and resilience (Sakinah, Farida, Azni, & Najwa , 2024)

By this research paper "Disaster Prep" is a mobile edutainment application aiming to improve catastrophe preparedness and resilience through interactive, gamified learning. The software includes 3D simulations, real-time notifications, and quiz-based challenges related to floods, earthquakes, and disease outbreaks. It contains English and Malay voiceovers, e-certificates for level completion, and an easy-to-use interface. Usability testing revealed improved knowledge, positive behavioral changes, and greater community resilience. The app passed all seven test cases, demonstrating its usefulness in engaging users and promoting emergency preparedness. Disaster Prep addresses the lack of awareness and practical understanding regarding disasters, especially in Malaysia, where floods and haze are regular. The app also promotes sustainability by lowering paper usage and increasing energy efficiency.

Inspired by Disaster Prep, my application will teach catastrophe preparedness through gamified, interactive learning. Because I do not have training modules, the app will focus on realistic simulations and practical scenarios to teach users about disaster risks and responses. The app will include voiceovers in different languages (for example, English, simple commands), as well as quiz-based challenges to evaluate users' knowledge. The software will also contain localized risk assessments and interactive guidelines for creating emergency kits and planning evacuations. My software will have uses

gamification to raise disaster awareness, encourage behavioral changes, and increase community resilience, particularly in disaster-prone areas. It will also enhance sustainability by minimizing dependency on physical resources and making it more accessible to diverse groups.

3.4.14 Managing flash flood crises with cultural perspectives: A user-centric feature identification study (Sumendra & Umar Ali, 2025)

This study “Managing flash flood crises with cultural perspectives”, investigates the role of cultural perspectives, using Grid-Group Cultural Theory, in developing user choices for an in-vehicle flash flood app. A survey of 351 respondents from flood-prone locations revealed four cultural biases individualistic, egalitarian, hierarchical, and fatalist that influence how users perceive and prioritize app features. Key findings show that consumers place a high value on real-time updates, GPS-guided maps, and locationsharing tools in emergency circumstances, while features such as hands-free voice commands and ringing mechanisms are regarded less beneficial. The study emphasizes the need of including cultural biases in app design to improve user adoption, trust, and disaster management efficacy.

Inspired by this research, I plan to create a disaster application that includes user-centric and culturally sensitive features. Specifically, I will prioritize real-time updates and alerts to provide users with timely information about flood-prone areas, interactive GPS-guided maps to assist users in navigating and avoiding flooded zones, and location-sharing functionality for emergencies, which allows users to share their location with responders and family members. In addition, I plan to employ community interaction tools like in-app chat and social network integration to encourage user collaboration and information sharing. By including these components, the application aims to improve user acceptance, trust, and efficacy in handling flash flood disasters, ensuring that it fits the needs of a diverse set of users.

3.4.15 Landslide and flood disaster: Causes and its responses (Tika Ram, 2024)

The study "Landslide and flood disaster", investigates the causes and human responses to landslides and floods, focusing on their destructive effects on lives, infrastructure, and economies. Landslides and floods are caused by geological, morphological, physical, and human factors such as unstable terrain, high rainfall, deforestation, and bad land management. Human responses range from traditional methods such as tree planting, bioengineering, and increasing house levels to more modern strategies such as catastrophe risk reduction frameworks and migration. The study emphasizes the need of combining local knowledge with scientific methods for efficient catastrophe management. Key findings indicate that, while natural factors play an important impact, human activities such as urbanization and poor land use increase disasters. Effective mitigation necessitates a mix of community-based adaptation, policy reform, and technical breakthroughs.

Inspired by this research, I intend to create a complete disaster management strategy that combines indigenous knowledge with modern technology.. To improve early warning systems, I will combine realtime data collecting and community participation, allowing for timely and accurate notifications. The plan aims to lessen the impact of landslides and floods by combining traditional practices with scientific breakthroughs, while also developing resilience and sustainability in vulnerable communities.

3.5 Summary of Literature Review (Findings and Critical Analysis)

This literature review/case study underlines the expanding importance of technology in disaster management, with apps emphasizing real-time warnings, offline capabilities, and community interaction. Many apps, such as iHanda and Disaster Prep, excel at offering preparedness tools like weather updates, evacuation maps, and gamified learning, but they sometimes miss key features such as mental health assistance, cultural diversity, and long-term success. Apps, for example, rarely address disaster survivor's mental health issues or take into account cultural biases that influence user choices, as evidenced by the flash flood app study. Misinformation, data privacy, and accessibility in rural or low-

connectivity locations are all issues that have yet to be addressed. While GPS-guided maps and offline caching are useful, they must be combined with user-centered design to ensure simplicity and accessibility. Future disaster related apps should take a balanced approach, integrating modern technology, cultural sensitivity, mental health services, and long-term solutions to properly satisfy the different demands of users and communities.

3.6 Justification

The Community Disaster Preparedness App's technology stack is created with reliable and modern tools to provide smooth performance, scalability, and accessibility during disasters. Flutter is the major mobile application framework due to its cross-platform features which allow the program to run on both Android and iOS. Dart, the programming language, ensures rapid development and responsive UI performance. Firebase is used to manage user data, incident reports and admin tasks.

Because the program is now totally Android-based, offline maps are implemented using a Flutter map tile caching library, allowing users to browse maps even when there is no internet connection. OpenStreetMap is used for geolocation and offline map features, which assist users in locating shelters and view emergency spots. Disaster alerts are coupled with API like GDACS to offer real-time notifications for earthquakes, severe weather, and other natural disaster.

To ensure data availability during network disruptions, SQLite is utilized to store vital information such as emergency contacts, recommendations, and saved locations on the device itself. The admin-based validation mechanism allows administrators to analyze, approve, or refuse incident reports filed by users, preventing misinformation in important situations. A user-centered design strategy is used to make the app easy, straightforward, and accessible, particularly in stressful emergency situations. The testing phase consists of Unit Testing for individual components, Integration Testing for data flow between modules, System Testing to evaluate overall functionality, User Acceptance Testing

(UAT) to validate real-world usability, and Performance and Security Testing to ensure the app remains stable, responsive, and safe even during high-load disaster scenarios.

Chapter 4: Primary Research

4.1 Objectives of Primary Research

The primary objective of this research is to gather first-hand information from users and stakeholders to validate the necessity and effectiveness of a disaster preparedness mobile application. The study aims to:

- Identify gaps in existing applications through market research.
- Understand user needs and expectations for disaster alert and emergency response applications.
- Analyze how users interact with the current solutions and what features they find most useful or missing.
- Collection of data from a survey questionnaire to support decision making in the development and customization of the application.

4.2 Market Research (Existing Products)

Existing App	Key Features	User Feedback (Good Aspects)	User Feedback (Issues Faced)

BIPAD Portal (Nepal, 2025)	Real-time disaster alerts, risk mapping, policy access	Users trust the data since its from official sources; helpful for researchers and policy planners.	Users feel its not userfriendly, no offline support or personalization options.
Pahiro Alert	Landslide and Flood early warnings, real-time weather tracking	Appreciated for scientific accuracy in flood and landslide alerts; useful in monsoon.	Users note that alerts are limited to certain disasters; not comprehensive in emergency support.
Nepal Red Cross First Aid App (Society, 2023)	First aid guides, disaster preparedness checklists	Great for learning basic first aid; helpful preparedness materials for families.	No real-time alerts, users miss interactive features like maps or live updates.
SAFER Nepal	Disaster safety training, emergency response tips	Community-focused, promotes awareness and safety drills in schools and communities.	Not useful during actual disasters; lack real-time updates and shelter navigation.

Table 4: Market Research of Existing Project's

1.3 Survey Questionnaire

1. What is your age group?

- Below 18
- 18-30

- 31-50
- Above 50

Objective: To understand the demographic distribution of potential users and tailor features accordingly.

2. What is your gender?

- Male
- Female
- Prefer not to say
- Other

Objective: To find out the gender of users so I can design the app to work well for everyone.

3. Which area do you live in?

- Urban
- Rural

Objective: To assess the different needs and challenges faced by urban versus rural communities.

4. Have you experienced a natural disaster (e.g., earthquake, flood, landslide)?

- Yes
- No

Objective: To gauge personal experience with disasters and the relevance of the app's features.

5. Are you aware of any mobile apps for disaster preparedness?

- Yes
- No

Objective: To find out if people already know about disaster preparedness apps, so I can decide how best to introduce my own app.

6. Do you currently receive any disaster alerts on your phone (via SMS or apps)?

- Yes
- No

Objective: To see if users already get alerts, helping me understand if there's a gap our app can fill.

**7. Which of the following disaster feature do you think are most important?
(Select those you think are important)**

- Real-time alerts
- Offline accessibility
- Shelter locator
- Emergency checklists
- First aid guides

Objective: To prioritize the features that users find most valuable for disaster readiness.

8. How frequently do you check for disaster alerts or updates?

- Frequently
- Occasionally
- Rarely
- Never

Objective: To analyze user awareness and frequency of disaster alert checks, which will help in optimizing notification delivery.

9. How often do you use mobile apps to get disaster information?

- Daily
- Weekly
- Monthly
- Rarely

Objective: To know user dependency on mobile apps for disaster information.

10. Would you prefer an app that works offline in case of network failure?

- Yes
- No

Objective: To assess the importance of offline accessibility due to potential internet connectivity issues.

11. How important is it for the app to include geolocation features (finding nearby shelters)?

- Very Important
- Somewhat Important
- Not Important

Objective: To determine user preferences for geolocation services and the necessity of incorporating shelter mapping into the app.

12. Do you find personalized safety checklists useful during emergencies?

- Yes
- No

Objective: To know the necessity of personalized safety checklists during emergencies.

13. What additional features would you like to see in a disaster preparedness app? Long – Paragraph

Objective: To gather suggestions for enhancing the app's functionality and user experience.

14. What challenges have you faced in receiving timely disaster alerts or emergency information? Long – Paragraph

Objective: To discover typical problems users have with disaster notifications, hence improving the app's alert system and reliability.

15. Would you be interested in testing this app when it becomes available?

- Yes
- No

Objective: To identify user interest.

16. Do you have any concerns about using mobile technology for receiving disaster alerts? Long – Paragraph

Objective: To understand potential barriers or privacy issues that need to be addressed.

17. Any additional comments or suggestion? Short – Paragraph

Objective: To improve the app through any further feedbacks.

4.4 Data Collection and Analysis

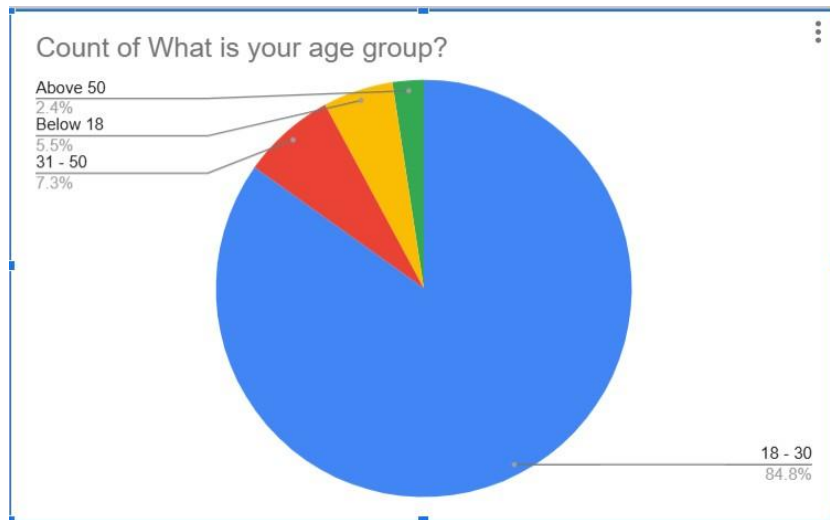


Figure 9: Pie Chart showing distribution of respondents by Age Group

The above pie-chart shows the distribution of respondents across different age groups. The majority of respondents is 84.8% which fall in the 18-30 age group. This indicates that the survey or study primarily captured a younger age groups. The 31-50 age group represents 7.3% of the respondents, while those below 18 and above 50 are for 5.5% and 2.4% respectively.

2. What is your gender?

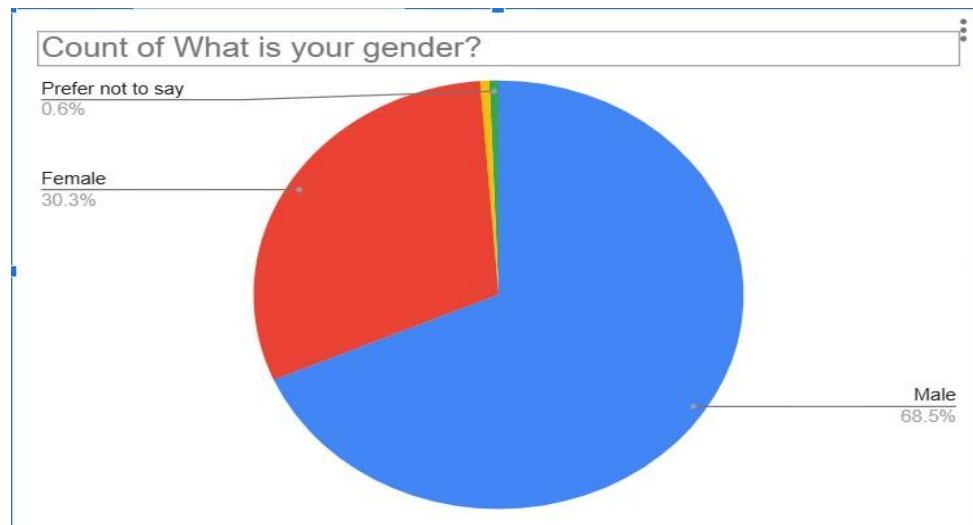


Figure 10: Pie Chart showing distribution of respondents by Gender

The above pie-chart shows the gender distribution of responders. A substantial majority, 68.5%, identify as male, with 30.3% identifying as female. A small fraction (0.6%) prefer not to reveal their gender. This distribution shows that male participants are overrepresented in the survey or study.

3. Which type of area do you live in?

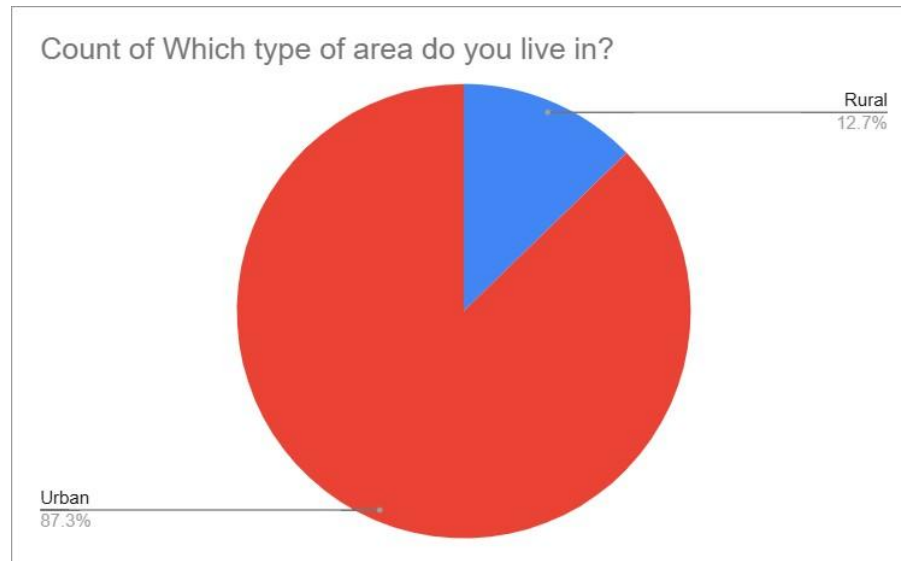


Figure 11: Pie Chart showing distribution of respondents by type of area they live in

The above pie-chart shows the type of area people live in. A high majority of people 87.3% of respondents live in urban areas where as only 12.7% live in rural areas. This indicates that most of the respondents living in urban areas participated more in the survey.

4. Have you experienced a natural disaster (e.g., earthquake, flood, landslide)?

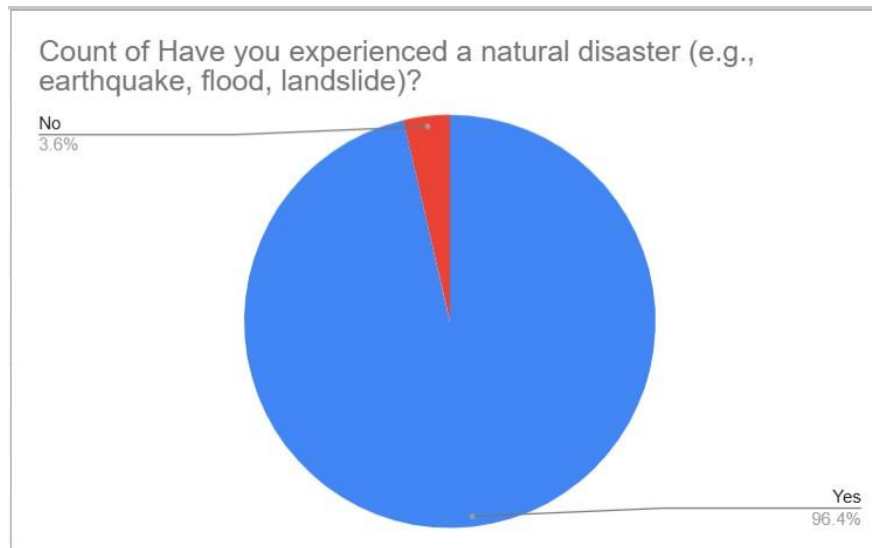


Figure 12: Pie Chart showing distribution of respondents by experience with natural disaster

The above pie chart shows the distribution of respondents based on their experience with natural disasters. The responses are divided into two categories: those who have and those who have not experienced a natural disaster. Majority of respondents have faced or experienced a natural disaster (96.4%) where only 3.6% of respondents have not experienced a natural disaster.

5. Are you aware of mobile apps for disaster preparedness?

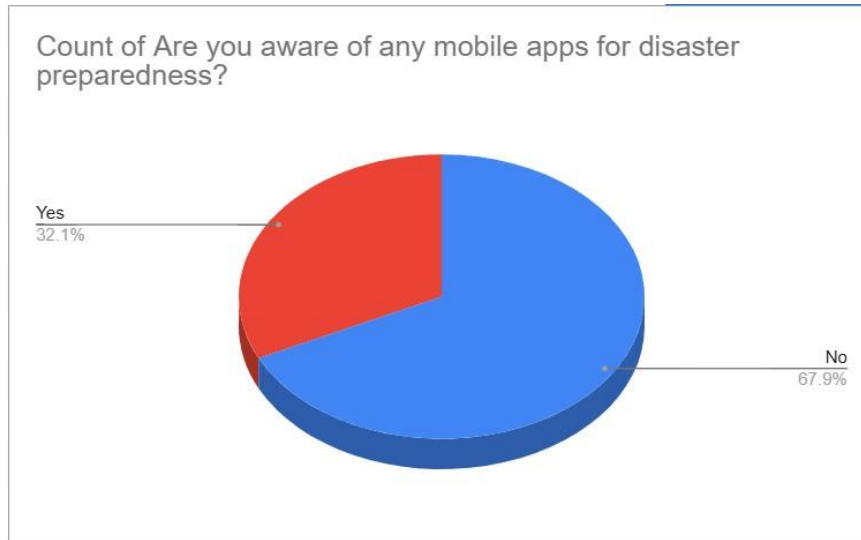


Figure 13: Pie Chart showing distribution of respondents by any awareness of mobile apps for disaster preparedness

The above pie-chart shows the distribution of respondents who are aware of mobile apps for disaster preparedness. Majority of people (67.9%) are not aware of mobile apps related to disaster preparedness which indicates respondents who had gone through the survey are not aware of it. Whereas, only 32.1% people are aware of mobile apps related to disaster preparedness.

6. Do you currently receive any disaster alerts on your phone (via SMS or apps)?

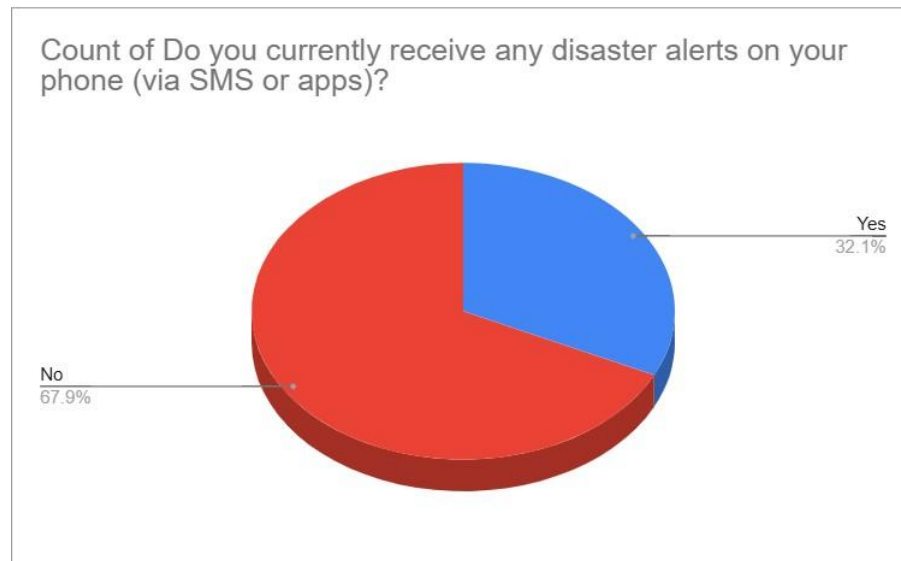


Figure 14: Pie Chart showing distribution of respondents by receiving any disaster alerts on phone

The above pie-chart shows the distribution of respondents by receiving any disaster alerts on phone. Majority people have not received any disaster alerts on their phone (67.9%) which indicates that the people do not receive any disaster alerts through SMS or any mediums. Only 32.1% of the respondents get the alerts.

**7. Which of the following disaster feature do you think are most important?
(Select those you think that are important)**

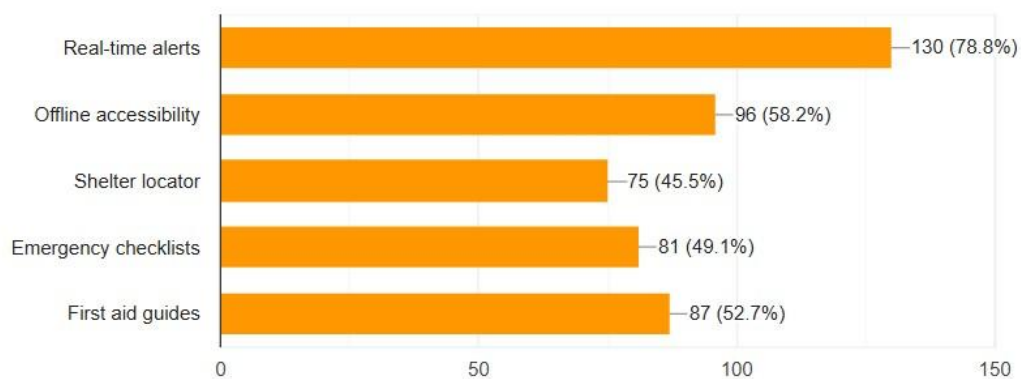


Figure 15: Bar Graph showing the distribution of respondents with the most important disaster feature

The bar graph shows how respondents ranked the significance of certain disaster characteristics. The most important features have the higher bars, and the features are prioritized according to the number of selections. This information aids in determining which features disaster preparedness apps should focus.

8. How frequently do you check for disaster alerts or updates?

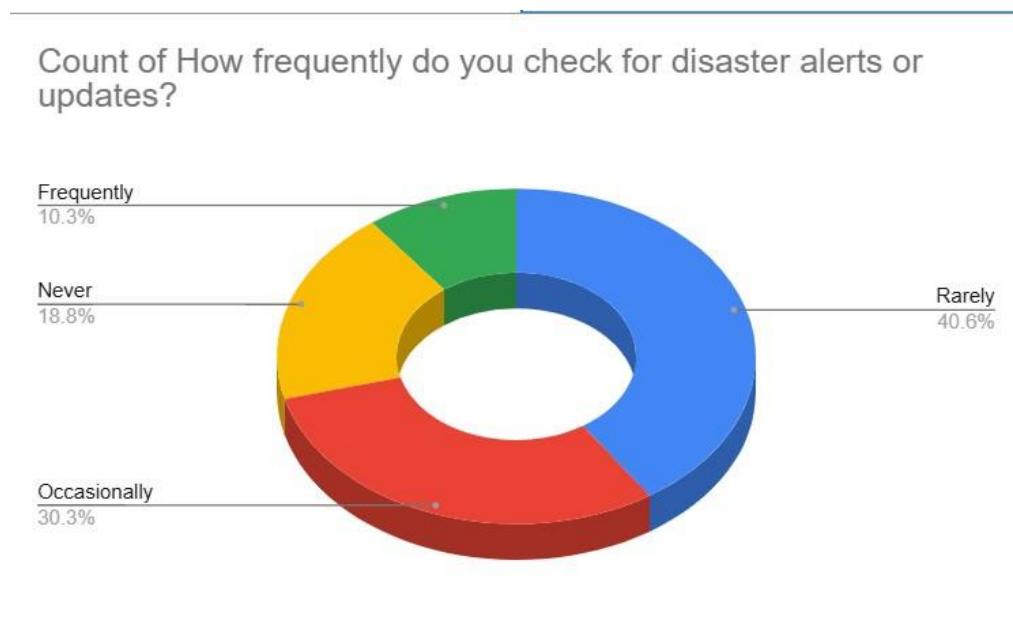


Figure 16: Pie Chart showing distribution of respondents by frequency of checking for disaster alerts or updates

The above pie-chart shows that which respondents look for disaster alerts or updates is can be seen in the pie chart. 30.3% of respondents check for notifications occasionally, and 40.6% of respondents check infrequently. Just 10.3% of respondents check regularly, while 18.8% never check for updates or crisis alerts. This suggests that a sizable percentage of respondents do not frequently interact with information on disasters.

9. How often do you use mobile apps to get disaster information?

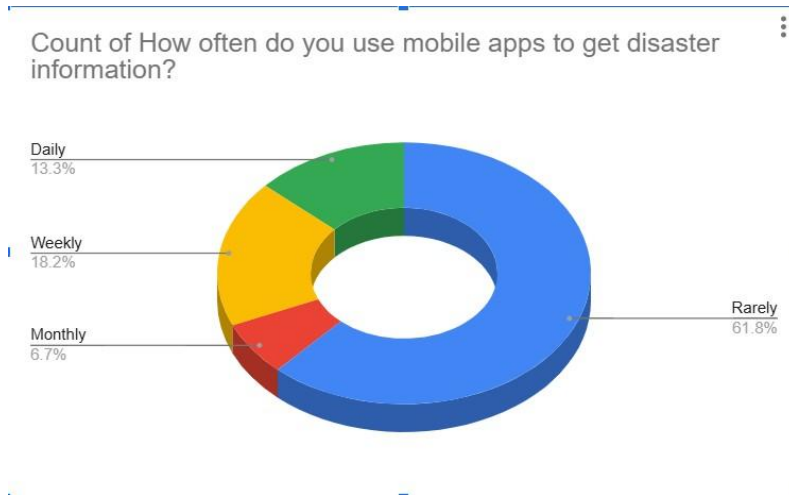


Figure 17: Pie Chart showing Usage frequency of mobile apps to get disaster information by respondents

The pie chart shows how often respondents use mobile apps to obtain disaster information. The majority of respondents, 61.8%, use these apps rarely, while 18.2% use them weekly. A smaller percentage, 13.3%, use the apps daily, and 6.7% use them monthly. This indicates that while a significant portion of respondents are not regularly using mobile apps for disaster information, there is still a notable percentage that engages with these apps on a weekly or daily basis.

10. Would you prefer an app that works offline in case of network failure?

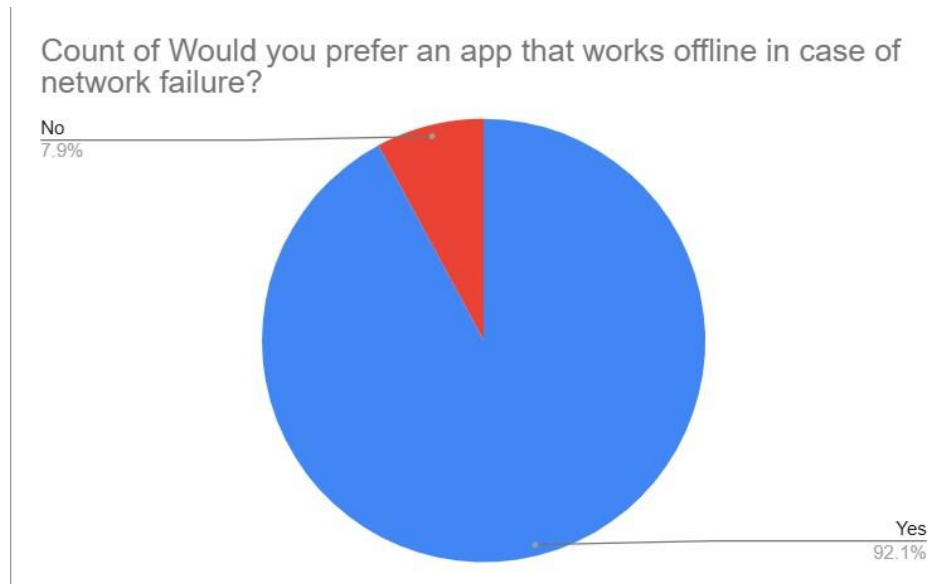


Figure 18: Pie Chart showing preference for offline functionality by respondents

The above pie-chart shows that almost all of the respondents wants offline functionality in this application which is 92.1% whereas only 7.9% people do not think the same.

11. How important is it for the app to include geolocation features (finding nearby shelter)?

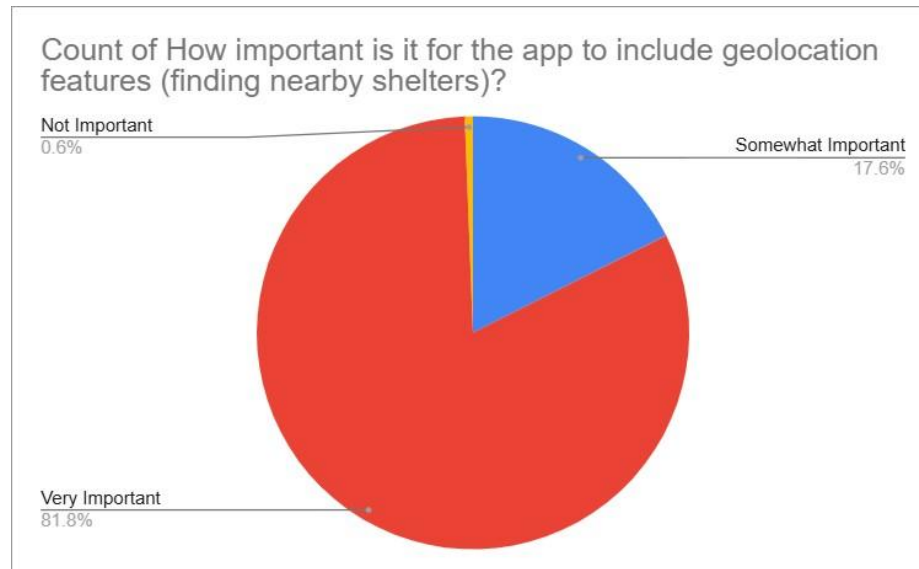


Figure 19: Pie Chart showing the importance of geolocation features by respondents

The above pie-chart shows that geolocation features is very important. Majority of people voted for 81.8% where 17.6% think it is somewhat important in this application. 0.6% think its not important feature in this disaster preparedness application.

12. Do you find personalized safety checklists useful during emergencies?

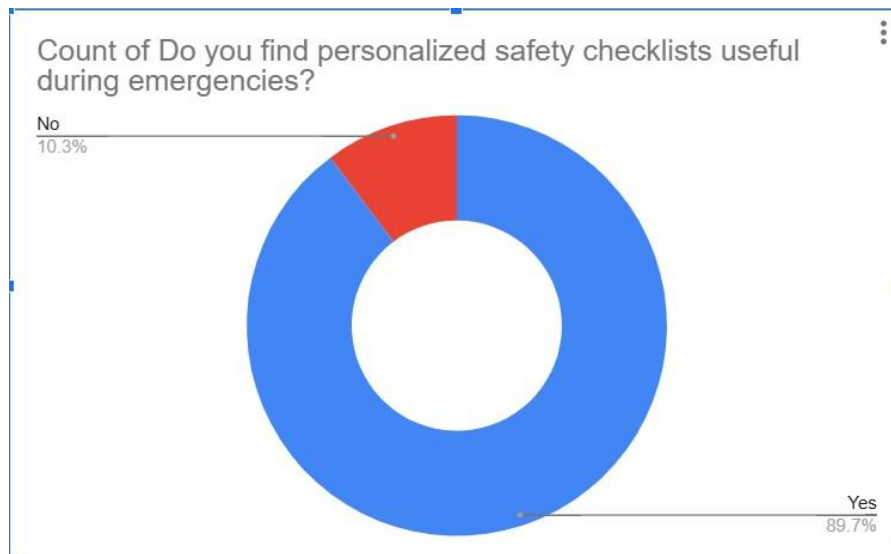


Figure 20: Pie Chart showing the usefulness of personalized safety checklists by respondents

The above pie-chart shows that 89.7% of the respondents feel that having personalized safety checklists is useful during emergencies where 10.3% of the respondents do not find it useful.

13. Would you be interested in testing this app when it becomes available?

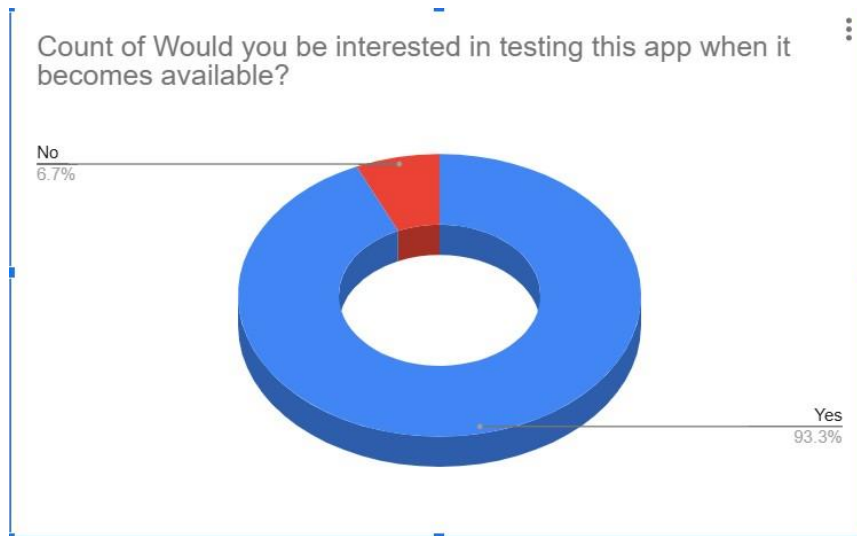


Figure 21: Pie Chart showing the interest in testing of this app by respondents

The above pie-chart shows that 93.3% of respondents want to test this app when it becomes available and only 6.7% don't want to test it. This indicates that respondents have high hopes for this application.

Chapter 5: Artefact Planning

5.1 Requirement Analysis

Requirement analysis is an important procedure that allows the success of a system or software project to be determined (geeksforgeeks, 2025).

5.1.1 Functional Requirements

S. No.	Requirements	Priority*
1	User must be able to login using email and password or Oauth 2.0 (Google)	MUST

2	User must receive real-time disaster alert via the GDACS API	MUST
3	User must access offline maps and see the shelters and emergency contacts	MUST
4	User must submit incidents reports with photos to avoid misinformation	MUST
5	User must be able to access the disaster preparedness checklist	MUST
6	User should be able to add the shelter locations on the map	SHOULD
7	Admin must view the incident reports and approve	MUST

Table 5: Functional Requirements

Functional requirements explain what the system should do, such as specific functions or tasks. It focuses on system behavior and features (geeksforgeeks, 2025).

5.1.2 Non-Functional Requirements

S. No.	Requirements	Priority*
1	System must provide fast response time for tasks like disaster alerts, map loading and offline functionalities.	MUST
2	The app should be compatible with both Android and iOS operating systems.	COULD
3	System must encrypt user data (AES-256) including user's personal information and login credentials.	MUST

4	System must support huge number of users during peak disasters.	SHOULD
5	System must ask user permission for using user current location	MUST
7	System must be user friendly to use	MUST

Table 6: Non-functional Requirements

Non-functional requirements explain how the system should function, including system properties and quality. It focuses on performance, usability, and other quality features (geeksforgeeks, 2025).

5.2 Software Requirements

S. No.	Requirements	Priority*
1	Operating System: Android and iOS	COULD
2	Database: Firebase (handling user data, incident reports and admin tasks)	MUST
3	Geolocation Services: Open Street Map	MUST
4	Authentication: Oauth 2.0 (Google)	SHOULD
5	Disaster Alert API: GDACS	MUST

Table 7: Software Requirements

Software requirements describe the intended system's features and functionalities. Requirements convey consumers' expectations of the software product. The needs may be evident or concealed, known or unknown, expected or unexpected from the client's perspective.

5.3 Hardware Requirement

Hardware requirements are the minimum specifications like as processor speed, memory, and disk space required to install an operating system like Windows. For maximum service and application performance, it is advised that your hardware exceeds these criteria (ScienceDirect, 2010).

5.3.1 Development

- Personal laptop or dekstop with minimum RAM of 8GB or higher and enough space for the development tools and emulator to run.
- Android Devices for testing the application.
- Flutter, Firebase, Dart and Android Studio/VS Code tools.
- Windows 10/11

5.3.2 Deployment

- Firebase for hosting on local server.
- GPS for location based services.
- Android phone with 4GB RAM or higher
- Minimum of at least 150-200 MB storage for application
- Network support of 4G and 5G

5.4 System Design

5.4.1 Use Case Diagram

A use case diagram summarizes details about a system and its users. It is typically displayed as a graphical representation of the interactions between various elements in a system (techtargert, 2020).

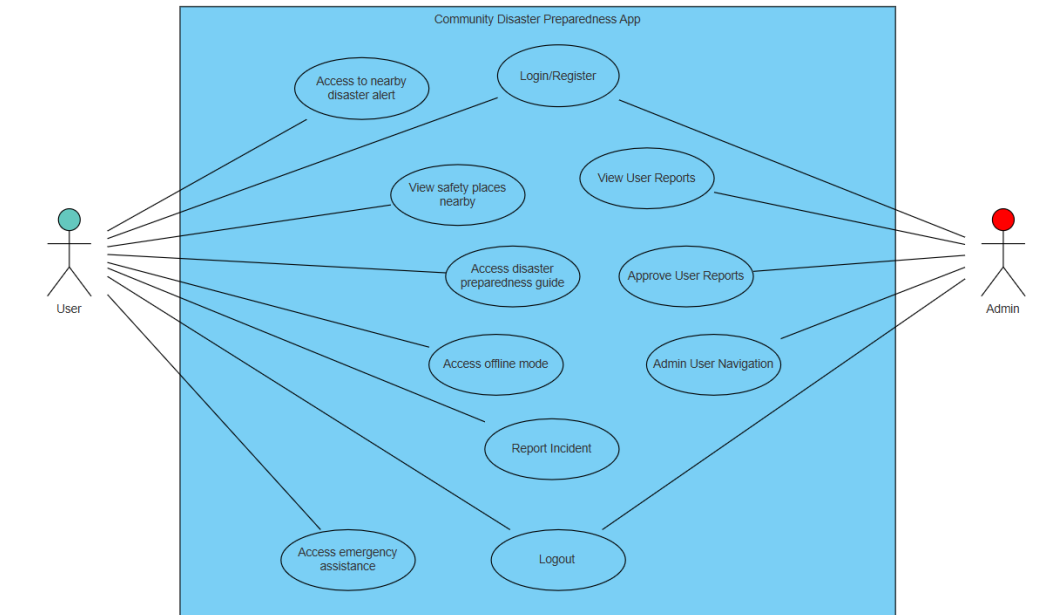


Figure 22: Use Case Diagram of Community Disaster Preparedness App

This Use Case Diagram shown in the figure illustrates that there are two actors (User and Admin) whose functionalities are different from each other in this application. Users can login or register to perform essential functions. They can receive real-time disaster alerts, view nearby disaster alerts, locate nearby shelters, and contact emergency services. Additionally, offline mode is also available with incidents reporting, and emergency checklist viewing to stay prepared.

System administrators are responsible for administering incident reports, and emergency resources (such as shelters and numbers). They guarantee that information is accurate and reliable before it is delivered to users.

5.4.2 Database Schema

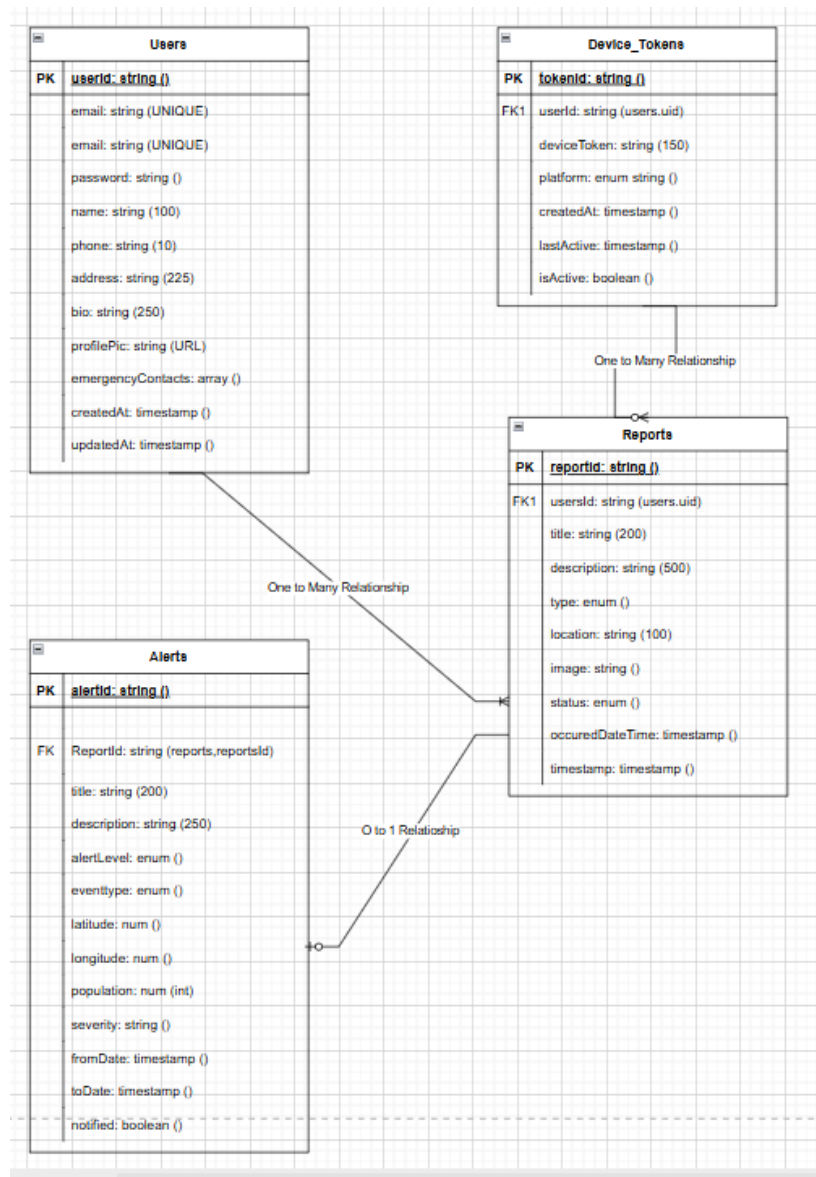


Figure 23: Entity Relationship Diagram (ERD) of Community Disaster Preparedness Application

Entities

1. Users

- Primary Key: uid (string)
- Attributes: address, bio, createdAt, email, emergencyContacts, name, password, phone, profilePic, updatedAt

2. Reports

- Primary Key: Not shown clearly, but likely automatically generated ID
- Attributes: description, image, location, occurredDateTime, status, timestamp, title, type, userID
- Foreign Key: userId → references users.uid

3. Alert

- Attribute: alertLevel, calculationType, country, dateAdded, dateModified, description, eventname, eventid, eventtype, fromDate, latitude, longitude, notified, population, severity, title, toDate

4. Device Tokens

- Stores devices tokens for push notifications.

Relationship Flow

1. User Submission:

- User (userid) submits a report, then the report is stored in reports collection with initial status: “Pending” and userId field links to the submitting user.

2. Admin Review:

- Admin views pending reports, then updates report.status from “Pending” to “Approved”

Chapter 6: Testing & Evaluation Plan

6.1 Testing Strategy/Plan & Test Cases

Software testing is the process of determining if a software application meets stated requirements and identifying any faults. The testing plan involves multiple methodologies to validate the app's functionality, performance, security, and usability (Taley, 2020).

6.1.1 Unit Testing

Unit Testing focuses on independently testing of individual components, functions and modules to ensure that they all function properly (Service, 2025). Unit tests for this application focused on authentication, user input validation, incident reporting, Firebase operations, map tile caching, etc. Unit testing contributes to clean architecture, simplifies debugging, and improves code quality by separating functionalities like “login validation”, “report creation”, and “saving offline map tiles.”

6.1.2 Integration Testing

Integration testing focuses on verifying the interactions and data exchange between different components or modules. (Patra, 2025). This testing phase ensures that data flows correctly between the frontend and the Firebase backend, that incident reports submitted by users reach the admin dashboard, that map and geolocation tools load effectively, and that API interactions like getting weather notifications are confirmed. The goal is to ensure that all components communicate without any errors or data conflicts.

Test Cases

Test Case No.	Test Case Description	Input	Desired Output
1.	Register with valid details	Enter full name, phone number, email, password and address	Login and show home screen
2.	Register with existing email	Use an email already registered	Error: The email address is already in use
3.	Login with invalid credentials	Enter incorrect email/password	Error: The auth credential is incorrect
4.	Login with valid credentials	Enter correct login details	Login successful
5.	Submit disaster report	Fill incident form and submit	Report successfully submitted
6.	Fetch incident reports in admin panel	Admin logs in and views reports	All users reports displayed
7.	Load offline map tiles	Open map in offline mode	Cached tiles load successfully
8.	Trigger notification permission request	App starts and requests notification access	User receives notification permission popup

Table 8: Integration Testing Test Cases

6.1.3 System Testing

System testing checks the entire and integrated system to ensure that it operates as intended across several platforms (Android and iOS) (IBM, 2025). It tests both functional and non-functional needs, including app responsiveness, offline support, error handling, map accuracy, notification delivery, and administrative functions. This step guarantees that the application functions reliably in real-world scenarios, such as several users reporting events at the same time.

Test Cases

Test Case No.	Test Case Description	Input	Desired Output
1.	Submit and approve disaster report	User submits report, admin reviews	Report shown to admin, admin can approve/reject
2.	Offline map loading	Open map without internet connection	Offline tiles load without errors
3.	Weather alert fetch	App requests weather data	Weather displayed
4.	Admin global reports view	Admin opens global reports	All approved reports displayed
5.	Deny notification permission	User denies notification access	No alert displayed
6.	App performance under load	Multiple reports submitted	App remains responsive and stable

Table 9: System Testing Test Cases

6.1.4 User Acceptance Testing (UAT)

Real users test the program in genuine circumstances to guarantee usability, clarity, and reliability during an emergency. Testers will attempt to submit reports, receive warnings, see offline maps, and navigate the user interface. Their feedback will help evaluate whether the technology is ready for deployment (Aakash, 2025).

Test Cases

Test Case No.	Test Case Description	Input	Desired Output
1.	Users submit an incident report	User fills form and submits	Report accepted and stored successfully
2.	Users check shelter locations	User views map in app	Accurate shelter locations displayed
3.	Users view offline map	Use app without internet connection	Offline map displayed
4.	User receive notification alerts	Disaster alert triggered	Notification reaches the device
5.	Users access to emergency contacts	Open profile and check emergency contacts	Information shown clearly when offline

Table 10: User Acceptance Testing (UAT) Test Cases

6.2 Test Evidence/ Screenshots

6.2.1 Splash Screen Loading

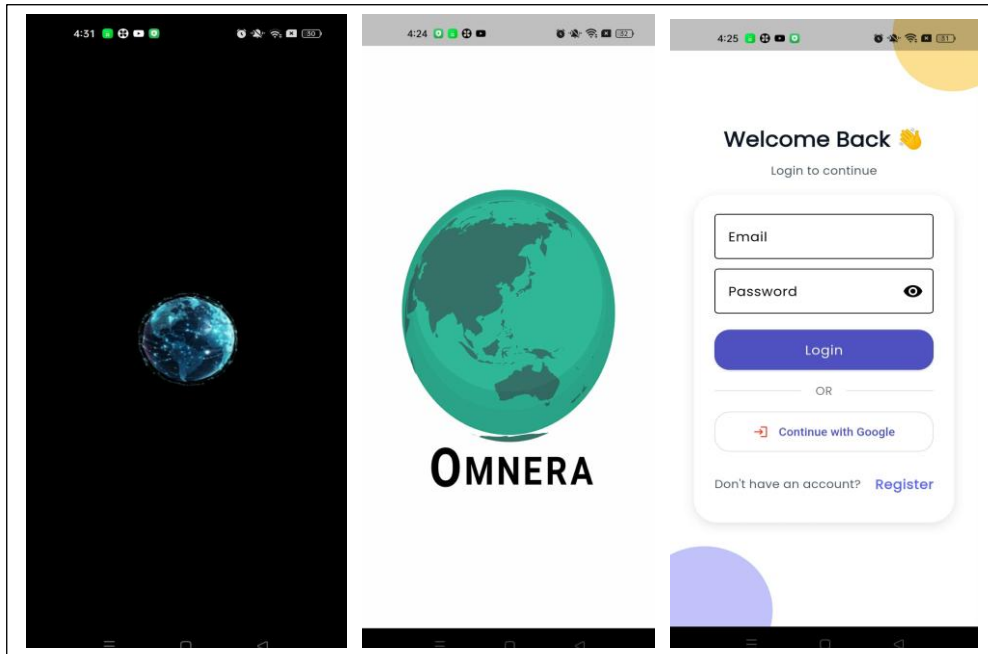


Figure 24: Splash Screen Loading

The above figure shows that the splash screen of Community Disaster Preparedness App (Omnera) loads correctly and transitions smoothly to the login page.

6.2.2 Notification Permission Popup

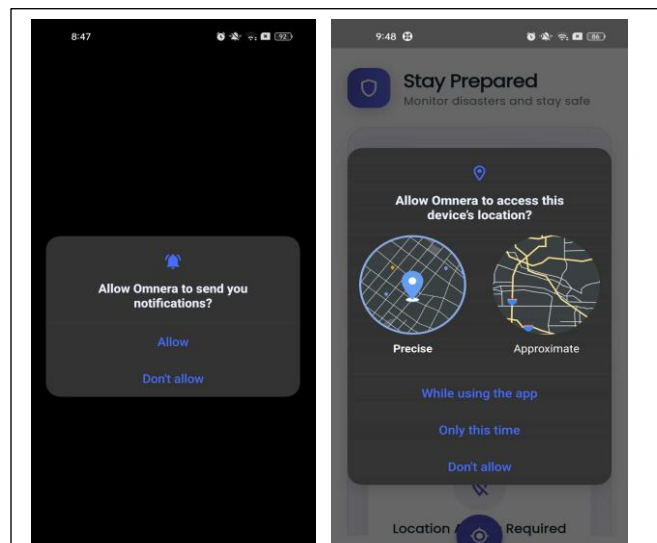


Figure 25: Notification permission popup on first launch

This is the proof that the app correctly requests notification permissions from users on the first launch.

6.2.3 Home Screen Display

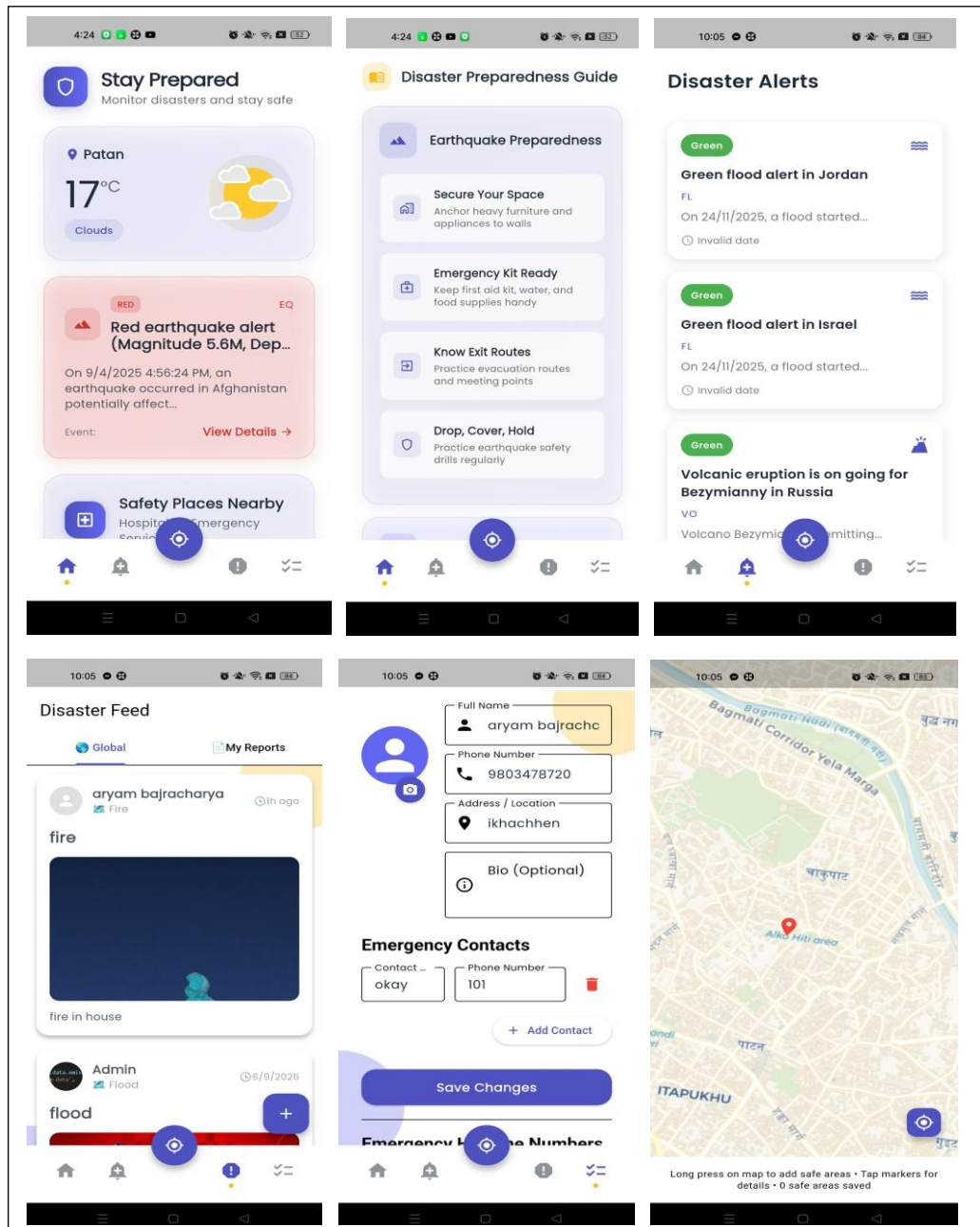


Figure 26: Home screen Display

The above figures demonstrates that the home screen loads all sections (alerts, map, report, profile, notifications,

6.2.4 Disaster Alert Notification (GDACS)

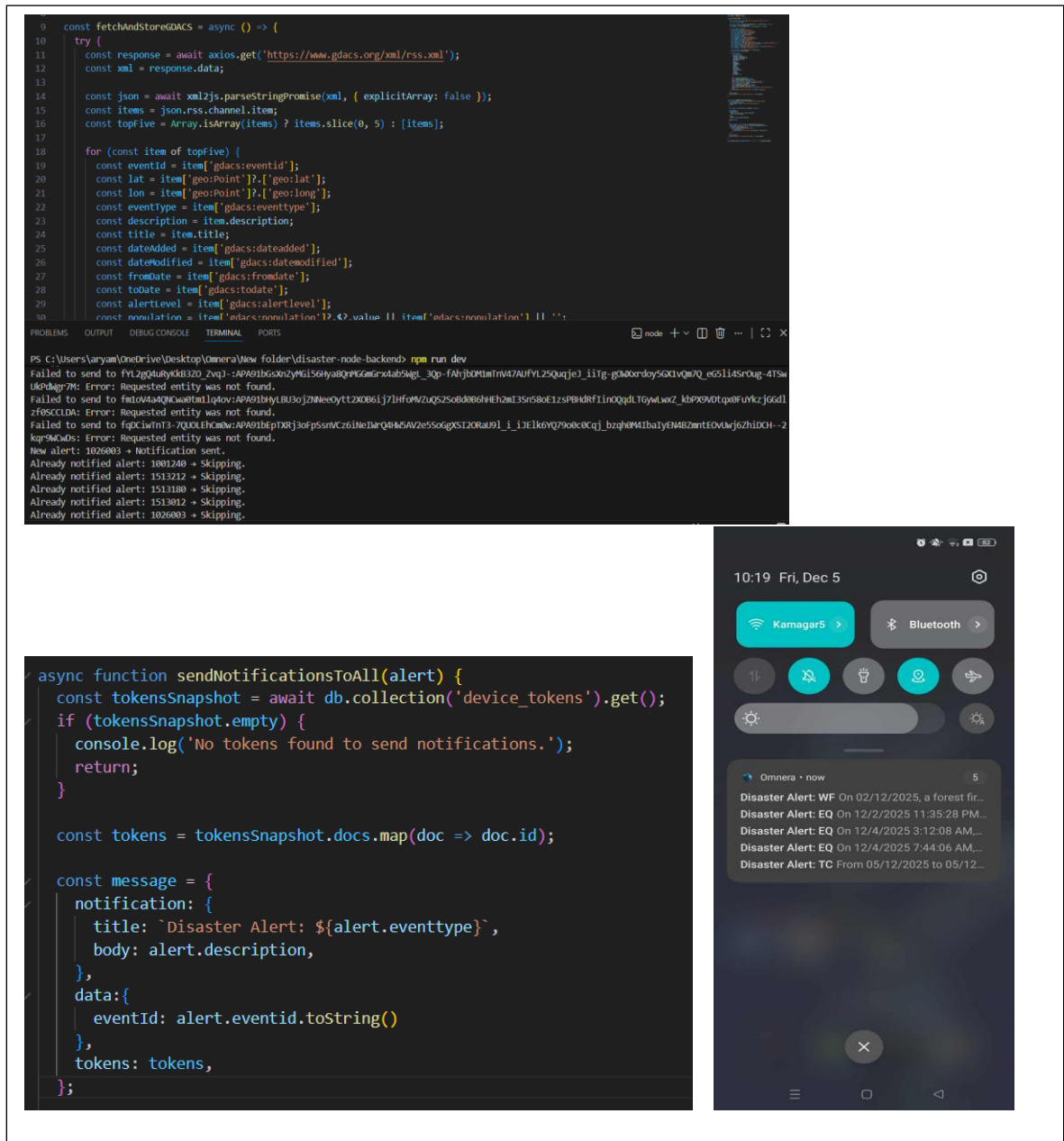
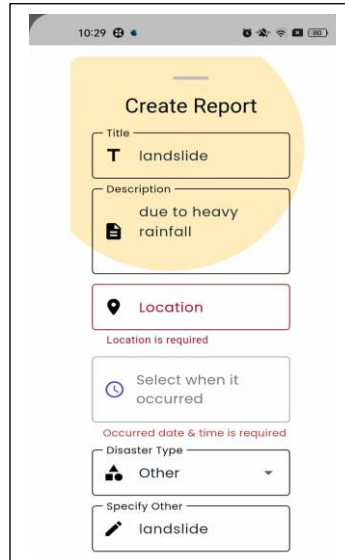


Figure 27: New Disaster Alert Notification Sent

This above figure is the evidence that real-time disaster alert from GDACS load correctly and sends the new notification otherwise skips.

6.2.5 Report Form Validation Error



The screenshot shows a mobile application interface for creating a report. The form is titled "Create Report" and contains several input fields. The "Title" field is filled with "landslide". The "Description" field is filled with "due to heavy rainfall". The "Location" field is empty, and a red error message "Location is required" is displayed below it. The "Select when it occurred" field is empty, and a red error message "Occurred date & time is required" is displayed below it. The "Disaster Type" field is a dropdown menu with "Other" selected. Below it, the "Specify Other" field is filled with "landslide".

Figure 28: Report submission error

This above figure shows that if the fields are left empty error message pops up and makes it unable to submit with incomplete data.

6.2.6 Successful Report Submission Popup

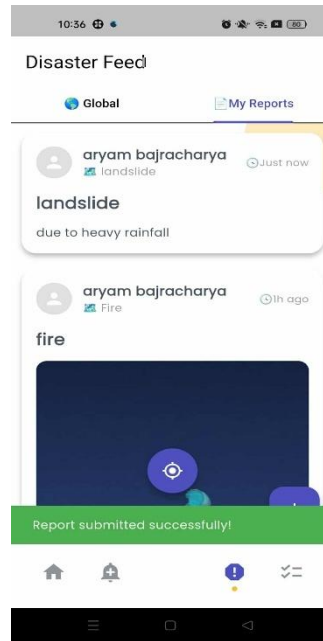


Figure 29: Report form submitted successfully

This confirms that the system shows a success message after the incident form is submitted with all the fields of the form are filled.

6.2.7 Firebase Database Entry

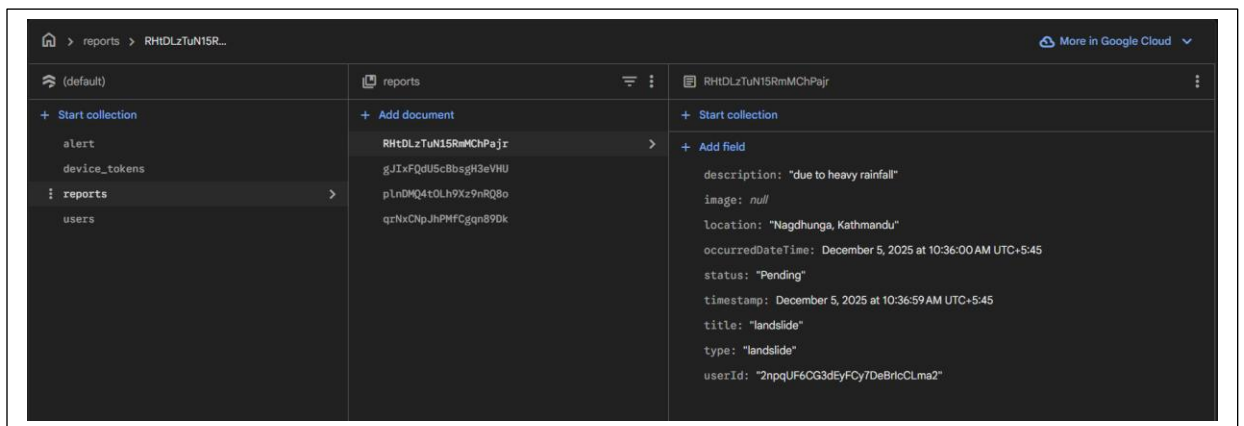


Figure 30: Firebase entry of the user report

This demonstrates that user reports appear correctly inside Firebase under the reports collection.

6.2.8 Admin Login Page

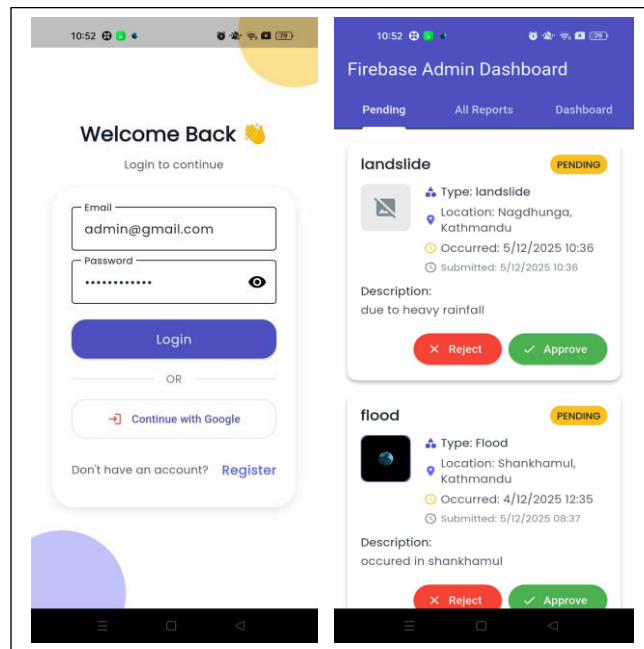


Figure 31: Admin Login Page

This shows the admin login interface.

6.2.9 Admin Dashboard Overview

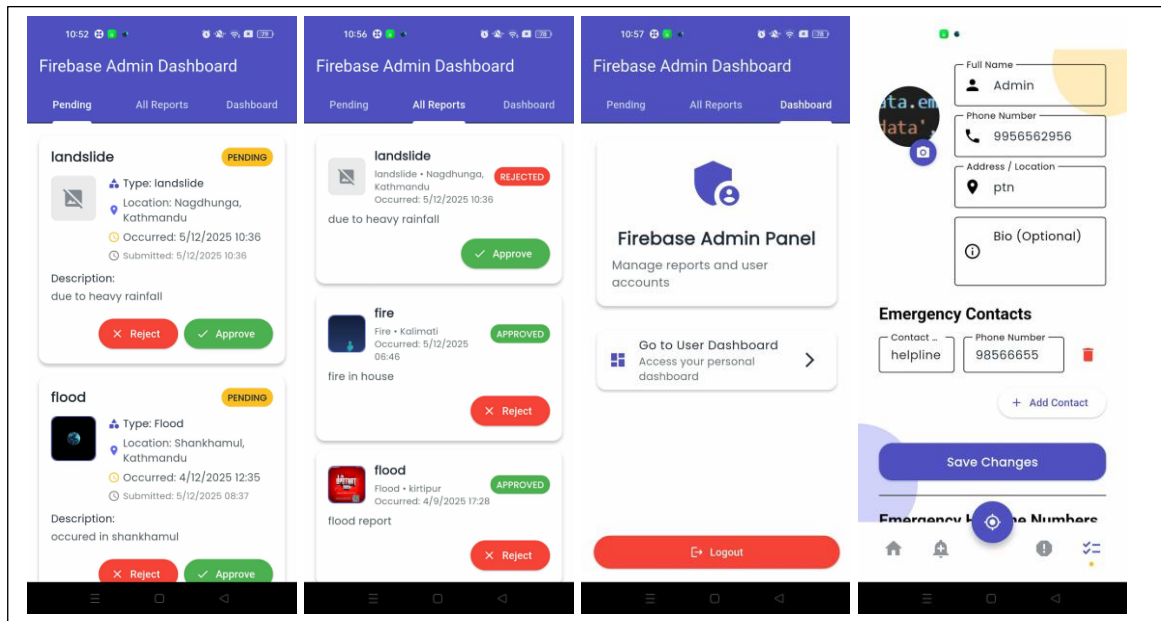


Figure 32: Admin Dashboard

This above figure shows the admin dashboard after login which shows the total number of reports, pending reports, approved reports and the profile section from which admin-user navigation is handled.

6.2.10 Offline Map Loading

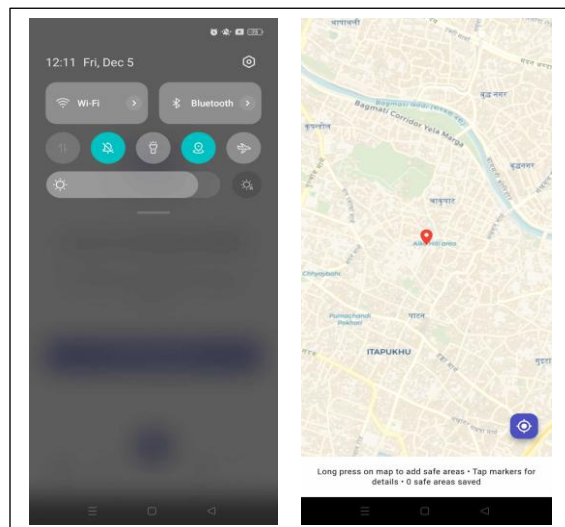


Figure 33: Offline Map Loading

This shows the offline map working without internet connection using cached tiles (flutter_map_tile_caching).

6.2.11 Shelter Location Marker

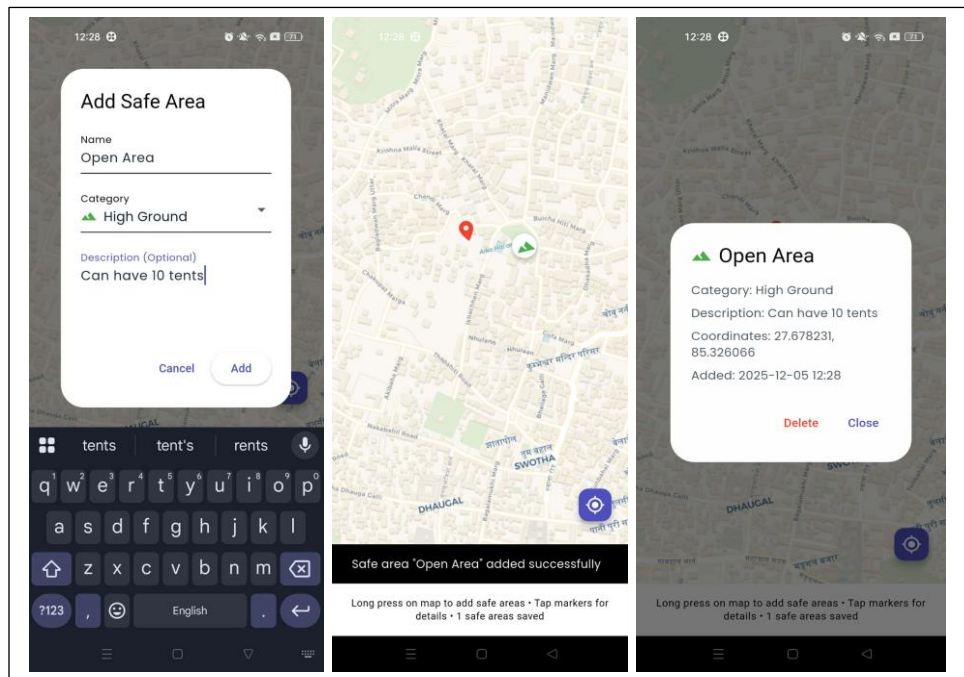


Figure 34: Shelter Location Marking

This above figure shows one of the main functions of this application. It displays shelter marking on the map with details when clicked which loads both online and offline.

6.3 Test Result Summary

The testing method was effective in building the Community Disaster Preparedness Mobile Application's functionality, performance, and usefulness. All important components, including authentication, incident reporting, disaster notifications, offline

mapping, and admin verification, worked as expected on Android smartphones. The majority of test case passed without major concerns.

A few issues were identified, including API key limits, slight delay in alert retrieval, and an inability to properly test on iOS due to the requirement for macOS. However, these problems have little impact on basic functionality.

Overall, the application is trustworthy, user-friendly, and dependable in real-world scenarios. It is ready for deployment though some changes are needed to be made in future versions.

6.4 Evaluation Strategy/ Plan

Software evaluation is the process of assessing a software application or system to determine its effectiveness, efficiency, quality, and suitability for a specific purpose or set of requirements (Loadium, 2024).

For this Community Disaster Preparedness Application, I will be implementing **DECIDE** framework. **DECIDE** framework is a systematic approach of evaluating systems and applications based on usability, functionality, and effectiveness. It consists of six steps: **Determine, Explore, Choose, Identify, Decide, and Evaluate.** (MUR, 2025)

1. D – Determine the Goals of the Evaluation:

- Evaluate how well the app enables users to receive real-time warnings, locate emergency shelters, and report incidents.
- Evaluate the app's usability under high-stress conditions.
- Identify potential concerns with alert accuracy, response speed and data security.

2. E – Explore the Questions:

- Have you experienced a natural disaster (e.g., earthquake, flood, landslide)?
- Do you currently receive any disaster alerts on your phone (via SMS or apps)?
- Would you prefer an app that works offline in case of network failure?

3. C – Choose the Evaluation Methods:

- User feedback collection will focus on evaluating usability together with dependability and utility through survey and feedback methods.
- The usability testing phase enables users to execute tasks such as shelter search and reporting incidents.
- The third method involves significant traffic simulation to confirm server performance as well as application stability.
- Security Assessment: Evaluate data encryption, authentication, and cyber threat preventions.

4. I – Identify the Practical Issue:

- **Timely and Accurate Alerts:** Users must receive real-time disaster notifications without any delays or inaccuracies.
- **Geolocation and Shelter Finding:** Users must be able to locate nearby emergency shelters quickly and reliably.
- **Offline Functionality:** This application should work without internet connectivity, as disaster may disrupt networks.
- **Efficient Report Verification:** Admins must be able to validate incident reports quickly to avoid misinformation.
- **Data Security and Privacy:** Personal data, location tracking, and reports should be secure from unauthorized access.

5. D – Decide the Ethical Issue:

- User privacy and confidentiality will be secured and respected.

- If disaster data is shared with government agencies, clear policies must be maintained to prevent misuse of user information.
- All users must receive equal and unbiased access to emergency resources.
- The app should be designed with strict data protection since the app collects sensitive information of the users.

6. E – Evaluate, Analyze and Interpret the Data

- Analyze success rate in task completion (e.g., how quickly users find shelters) □ Identify typical usability concerns and improve the UI/UX.
- Assess system reliability under stress and maximize performance.
- Ensure compliance with security best practices and address vulnerabilities.

Conclusion

Using the **DECIDE** framework, we can systematically test and improve the Community Disaster

Preparedness App's usability, dependability, and security. This evaluation assures that the software is functional in real-world environments and that users have a smooth, trustworthy experience.

Chapter 7: Development Process

The Community Disaster Preparedness Mobile Application was created through a structured, multi-phase process that included user registration and authentication, incident reporting, map integration, offline data caching, notification handling, and administration validation. This application was built with Flutter, Firebase was used for user report verification. The map feature was implemented, a library `flutter_map_tile_caching` was used for offline accessibility. The whole development process, including data handling, admin validation, and offline map administration, is outlined below.

7.1 Methodology

This project used the Agile methodology, which allowed iterative development and flexibility in solving technical difficulties like map caching, real-time notifications, and offline capabilities. The methodology allowed continuous feedback integration, and refinement of user and admin functionality. Regular testing and evaluation helped to ensure quality throughout the development cycle. (Alliance, 2001)

7.2 Tech Stack Utilized

Hardware: Android mobile devices (low-to-mid-end devices tested for offline map performance)

Software:

- Flutter (Mobile Application Framework)
- Firebase (Authentication, Firestore Database, and Push Notification)
- Node.js (Backend API for administrative validation and global reports)
- Flutter_map_tile_caching (Offline map caching for shelter and route access)
- Dart (Programming Language)

7.3 Development Planning and Data Collection

The project planning focused on defining all software features, prioritizing important disaster-preparedness capabilities, and breaking down activities into smaller, more manageable chunks. Disaster data and shelter locations were gathered from publicly available sources, including GDACS and municipal governments. User data fields, emergency contact information, and predetermined checklists were also gathered to ensure accurate functionality. Offline map tiles were preloaded to ensure usability during network outages, and preliminary tests were carried out on a variety of map providers (OpenStreetMap, Mapbox, HERE Maps), however several crashed or did not handle offline caching adequately; `flutter_map_tile_caching` proved stable.

7.4 Managing Android Permissions and Project Initialization

Managing Android permissions was an important aspect of the development process because various components of the Community Disaster Preparedness mobile app require system-level access. The program needs internet access for API interactions, map loading, and notification delivery. Location access (both fine and coarse) is required to establish the user's current location during an emergency, list local shelters, and compute distances offline. Additionally, **RECEIVE_BOOT_COMPLETED** is used to guarantee that notification listeners continue to function even after the device restarts. All of these permissions were defined in the `AndroidManifest` file, and runtime prompts were handled using the Flutter **permission_handler** package.

The project's startup included setting up the Flutter environment, integrating Firebase for push notifications, and configuring map-related packages for online and offline tile rendering. The `AndroidManifest` also contains the Firebase Messaging service and the default notification channel metadata, allowing emergency warnings to be delivered even when the app is not active. After initializing the project, all essential dependencies, such

as **firebase_messaging**, **flutter_map**, **flutter_map_tile_caching**, and **geolocator**, were added and configured to ensure proper application functionality.

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android" xmlns:tools="ht
  <uses-permission android:name="android.permission.INTERNET"/>
  <uses-permission android:name="android.permission.RECEIVE_BOOT_COMPLETED"/>
  <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
  <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

Figure 35: Code snippet of permission request at runtime

7.5 Firebase Integration and Admin Functionality

- Firebase was utilized to handle
- User Registration and Login
- Incident report submission and storage
- Push notifications for disaster alerts
- Admin verification of user reports

Admins can log in, review incident reports made by users, approve or refuse reports, view aggregated global reports, manage their profile, and log out.

Firebase's real-time features meant that report statuses were updated immediately and notifications were delivered reliably.

7.6 Offline Map Management and Geolocation

The offline map capability was developed using **flutter_map_tile_caching**. Shelters, emergency routes, and key places were downloaded and cached to provide access even

during network disruptions. Geolocation services were used to determine the user's location, which allowed them to find local shelters and resources. Real-time disaster notifications were originally planned for many APIs (GDACS, DHM Nepal, and OpenWeather), but owing to MOU limits, only GDACS was included in this version.

```
import 'package:flutter/material.dart';  
import 'package:flutter_map/flutter_map.dart';  
import 'package:flutter_map_tile_caching/flutter_map_tile_caching.dart';  
import 'package:geolocator/geolocator.dart';  
import 'package:latlong2/latlong.dart';  
import 'package:flutter_secure_storage/flutter_secure_storage.dart';  
import 'dart:convert';
```

Figure 36: Imports used for the map caching

```
class _MapPageState extends State<MapPage> {  
  final _tileProvider = FMTCTileProvider(  
    stores: const {'mapStore': BrowseStoreStrategy.readUpdateCreate},  
  );
```

Figure 37: Offline tile provider using FMTC

This creates an offline tile cache folder.

```
Future<void> _initializeMap() async {  
  await _getCurrentLocation();  
  await _loadSafeAreas();  
  setState(() {  
    _isLoading = false;  
  });  
}
```

Figure 38: Initialize offline map tile caching

```
import 'package:geolocator/geolocator.dart';
```

Figure 39: Import package for geolocator

```
Future<void> _getCurrentLocation() async {  
  try {  
    bool serviceEnabled = await Geolocator.isLocationServiceEnabled();  
    if (!serviceEnabled) {  
      _setDefaultLocation();  
      return;  
    }  
  
    LocationPermission permission = await Geolocator.checkPermission();  
    if (permission == LocationPermission.denied) {...}  
  
    if (permission == LocationPermission.deniedForever) {...}  
  
    Position position = await Geolocator.getCurrentPosition(  
      desiredAccuracy: LocationAccuracy.high,  
    );  
  
    _currentLocation = LatLng(position.latitude, position.longitude);  
  } catch (e) {  
    _setDefaultLocation();  
  }  
}  
  
void _setDefaultLocation() {  
  _currentLocation = const LatLng(27.7172, 85.3240);  
}
```

Figure 40: Code snippets of Geolocator

First checks if the GPS service is on then checks the permission, if permission is denied it will print “location permission permanently denied”. Then tracks our location using latitude and longitude.

7.7 Notification Handling

To ensure that users receive disaster alerts on time, the application has a strong alerting system. Upon the first opening of the app, users are asked for the notification permission, which allows the system to send high-priority alerts even when the app is running in the background. Once permission is granted, the device's FCM token is safely saved in Firebase for later notification delivery.

Notification are sent automatically whenever fresh disaster warnings are received from the GDACS API. The server analyzes incoming GDACS data at certain times and sends push notifications to all registered devices.

```
const fetchAndStoreGDACS = async () => {  
  try {  
    const response = await axios.get('https://www.gdacs.org/xml/rss.xml');  
    const xml = response.data;  
  
    const json = await xml2js.parseStringPromise(xml, { explicitArray: false });  
    const items = json.rss.channel.item;  
    const topFive = Array.isArray(items) ? items.slice(0, 5) : [items];
```

Figure 41: Fetch disaster alerts from GDACS API

This part runs every 10 minutes (cron job), pulls the latest disaster alerts, and converts them to JSON.

```
await alertRef.set(alertDoc);  
await sendNotificationsToAll(alertDoc);  
await alertRef.update({ notified: true });
```

Figure 42: Saves new alerts to Firebase

If the alert does not exist, this creates it and if it exists but does not get notified. If the alerts are notified it skips sending which prevents duplicate notification.

```
async function sendNotificationsToAll(alert) {  
  const tokensSnapshot = await db.collection('device_tokens').get();  
  if (tokensSnapshot.empty) {  
    console.log('No tokens found to send notifications.');    return;  
  }  
  
  const tokens = tokensSnapshot.docs.map(doc => doc.id);  
  
  const message = {  
    notification: {  
      title: `Disaster Alert: ${alert.eventtype}`,  
      body: alert.description,  
    },  
    data: {  
      eventId: alert.eventid.toString()  
    },  
    tokens: tokens,  
  };  
};
```

Figure 43: Send push notifications to all devices using Firebase Cloud Messaging (FCM)

This is the actual logic for sending push notifications to all devices.

7.8 Frontend Display

Flutter was used to complete the frontend. The app consists of two screens: User and Admin.

- User Screen: Home, Map, Incident Report Form, Notifications, Profile
- Admin Screen: Dashboard, Incident Reports, Global Reports, Profile

The user interface (UI) was meant to be clear and simple to use, allowing even users with less technical background to properly use this app (tutorialspoint, 2025). Offline map tiles, cached data and push notification are smoothly integrated into the interface.

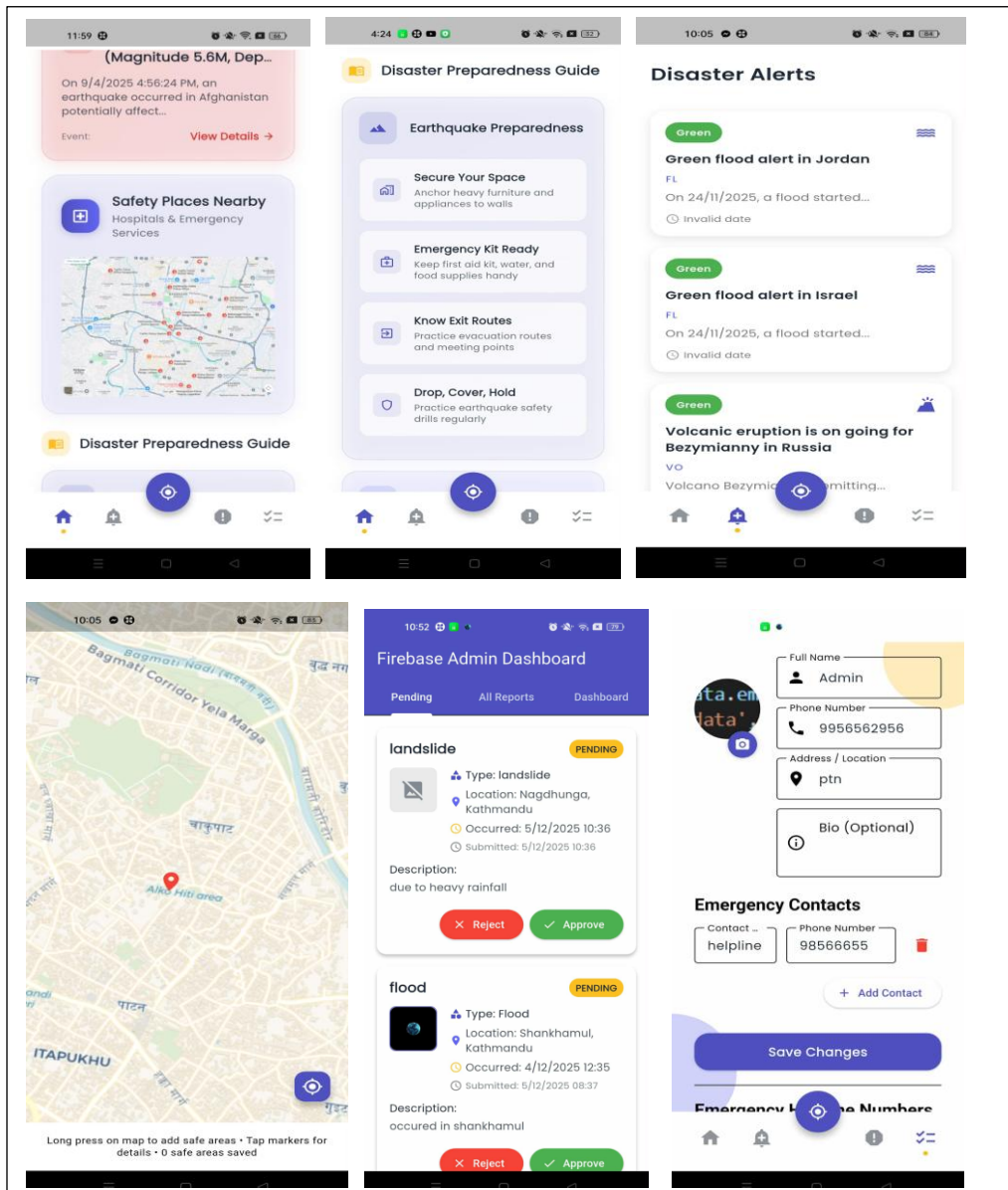


Figure 44: Frontend Display of Community Disaster Application

7.9 Backend Integration

The backend of the Community Disaster Preparedness Application was created with Node.js, Express.js, the Firebase Admin SDK, and scheduled Cron jobs. It serves as the primary layer that connects the Flutter mobile app to external disaster data sources and Firebase services. The system automatically gets global catastrophe alerts from GDACS, translates the XML feed to JSON, extracts critical information such event ID, disaster type, severity, and coordinates, and stores it in Firebase Firestore. To avoid repeating alerts, each stored event has a notified flag, which guarantees that push notifications are only delivered once per disaster. (w3schools, 2025)

Once a new alert is saved, the backend immediately delivers push alerts to all registered devices via Firebase Cloud Messaging. The device_tokens collection stores device tokens, which allow for effective multicast distribution and ensure that users receive warnings even when the app is closed. A Cron task refreshes data every 10 minutes, allowing for seamless automation and keeping the application up to date with real-time worldwide disaster information. (CrontabGuru, 2025)

```
const serviceAccount = {
  type: "service_account",
  project_id: process.env.FIREBASE_PROJECT_ID,
  private_key_id: process.env.FIREBASE_PRIVATE_KEY_ID,
  private_key: process.env.FIREBASE_PRIVATE_KEY.replace(/\\n/g, '\n'),
  client_email: process.env.FIREBASE_CLIENT_EMAIL,
  client_id: process.env.FIREBASE_CLIENT_ID,
  auth_uri: "https://accounts.google.com/o/oauth2/auth",
  token_uri: "https://oauth2.googleapis.com/token",
  auth_provider_x509_cert_url: "https://www.googleapis.com/oauth2/v1/certs",
  client_x509_cert_url: process.env.FIREBASE_CLIENT_X509_CERT_URL
};
credential = admin.credential.cert(serviceAccount);
} else {
  throw new Error('Firebase credentials not found. Please provide serviceAccountKey.json or environment variables.');
```

```
admin.initializeApp({
  credential: credential,
  projectId: process.env.FIREBASE_PROJECT_ID
});
```

Figure 45: Firebase Admin Initialization

```
cron.schedule(process.env.CRON_SCHEDULE || '*/10 * * * *', fetchAndStoreGDACS);
```

Figure 46: Cron Job Setup

Chapter 8: Evaluation

This chapter provides an overall evaluation of Community Disaster Preparedness Mobile Application based on user comments, performance behavior, and dependability.

Evaluation is critical in establishing whether a system is user-friendly, functional and reliable for real world use case scenarios specially in emergency situations. Throughout this stage, feedback was collected to identify the application's strengths and areas for development before its final release. (Loadium, 2024)

8.1 User Evaluation

To understand if the application is effective and usable for the disaster preparedness, a group of friends were given the apk file of the app. Each of them interacted with one features such as disaster alert, map, emergency contacts, shelter locator, and offline functions. Users shared both positive and place for improvements remarks based on their experience. Their feedbacks along with average rating were summarized in the following table.

S.No.	Criteria	Remarks	Overall Rating (Out of 5)
1.	UI/UX Design	• Clean and simple interface, easy to use. • Icons and color are easy to understand • Map panel is usable, but markers can overlap when many alerts appear • A dark/light mode would have been more good	4.5
2.	User Experience (UX)	• Alert feeds load smoothly but new notification should be there • Some users noted slight lag when the map loads many markers	4

		• Friends with no technical background were also able to use this app	
3.	Geolocation and Offline Access	• The location were exact of the users • Location-based services are limited without internet • Could include more offline resources such as evacuation maps • Some delays occur in low signal environments	3.5
4.	Disaster Detection & Alert (GDACS Integrgration)	• Notification appear even when the app is in background • Alert details are showing properly • User requested the option to	4.5

		filter alert types (earthquake, flood, landslide)	
5.	Authentication System	- Secure login/register with Firebase authentication - No user reported login failures - UI for login/register could be made more modern	4.5
6.	Reporting Incident	- Form should had camera option too - Reporting incident form had details part which was good	4

Table 11: User Evaluation of Community Disaster Preparedness Application

Chapter 9: Critical Analysis

The Community Disaster Preparedness project was created using an agile methodology that separated the project into requirement analysis, system development, and testing stages. The major goal of this project is to develop a reliable system that allows user to

get timely alerts, report disaster situations, geolocation based, and accessible preparedness guideliness to ensure people can act quickly during disaster. The system integrates real-time disaster notification from GDACS API, processes alert through a Node.js backend and delivers push notifications to user using Firebase Cloud Messaging (FCM). Along with alerts, the application provides offline access with interactive maps, emergency contacts and friendly navigation aimed to help people more effectively to natural disasters.

The project's early stages focused on analyzing growing demand for a dependable disaster-based information systems, particularly in areas where disasters occur frequently like earthquakes, floods, and landslides. A detailed literature analysis confirmed the necessity for a centralized disaster preparedness platform that combines real-time data with user-friendly information. Following this research, the features were finalized, which included real-time alerts, shelter locating in map, offline functionality, and etc. These requirements guided the design of the system architecture, wireframes, and backend workflow.

During the development process, several technical problems were faced in the areas of backend, integrating GDACS RSS feeds necessitated translating complex XML structures into readable JSON representation, which had to be extensively reviewed to avoid inaccurate data parsing. Additionally, executing scheduled cron jobs for periodic updates caused configuration issues during early testing. Another important difficulty was Firebase Cloud Messaging, where unstable device tokens and misconfigured server keys originally resulted in notification delivery errors. These difficulties were rectified by debugging, rearranging Firestore collections, and improving error handling in the Node.js backend.

On the mobile side, factors such as device technology and network strength influenced geolocation accuracy, resulting in delays while getting user locations or presenting multiple catastrophe flags. The application presently only supports Android, restricting its reach, and functionality like as map interactivity and real-time warning updates continue to require an active internet connection, which can reduce usability during network

failures commonly associated with catastrophic occurrences. Despite these restrictions, the system runs efficiently on most devices and gives offline access to vital preparedness guidelines, guaranteeing that users may access critical information even when there is no internet connection.

The program also has some significant advantages, such as a dependable automated alarm system backed by GDACS data, quick push alerts via Firebase, and a clean, straightforward design ideal for a wide spectrum of users. Moving forward, the system can be considerably improved by adding iOS support, integrating local government APIs, providing multilingual materials, and improving offline capabilities. Features such as risk-zone heatmaps, evacuation routes, AI-based predictions, and user-generated hazard reporting can help to increase situational awareness. Overall, the app achieves its primary goal of providing timely disaster alerts and preparedness information, and it has great potential for future expansion into a more comprehensive emergency-response platform.

Chapter 10: Limitation and Future Improvements

10.1 Limitations

The development of a real-time disaster preparedness and alerting system involved numerous of technical issues, including geolocation accuracy, live data sharing, mobile device restrictions, and other API requirements. Several limits were recognized during the development that have an impact on the application's performance and usability. The main limitations are described below:

1. Platform Restriction (Android Only)
 - The current version of the application is available on Android. Since the system relies heavily on Flutter plugins and Firebase services optimized for

Android, iOS users cannot access the app. This limits the overall outreach and reduces the diversity of user feedback.

2. Dependency on Internet Connectivity

- While disaster guidelines and maps are available offline, core features such as real-time interactions, and receiving alerts require a stable internet connection. During natural disasters when networks often fail these features may become temporarily inaccessible.

3. Performance Limitations on Low-End Devices

- Multiple disaster icons on the map or loading huge data feeds can cause latencies on low-cost handsets. Although tailored for modern hardware, the processing burden rises as the disaster dataset grows, occasionally compromising map smoothness and UI responsiveness.

4. Notification Permission Issues

- Push notifications rely on user. If the user denies the notification access, they will not get disaster alerts. This lowers overall system effectiveness, particularly during emergencies.

5. Dependency on External APIs (GDACS)

- The application relies on GDACS for worldwide data. If GDACS updates are delayed, service is interrupted, or the XML structure of the alerts changes, the backedn may fail to process or store them correctly. This reliance creates a risk outside the developer's control.

10.2 Future Scope

Despite its limitations, the Community Disaster Preparedness Application offers substantial room for growth and improvement. It has the potential to become a comprehensive disaster-response ecosystem by incorporating new technologies, improving user experience, and increasing accessibility. The important areas for future development are as follows:

1. Cross-platform Expansion (iOS Support)
 - Implementing platform-specific plugins and tailoring the system to iOS devices can significantly expand the application's reach. This would allow both Android and iOS users to benefit from the real-time alerting and readiness capabilities.
2. Integration of Local Government and Emergency APIs
 - Adding compatibility for national disaster management systems, municipal alert channels, and local early-warning APIs allows for extremely relevant, region-specific notifications. This would improve accuracy and help to close the information gap during local emergencies.
3. AI and Machine Learning Enhancements
 - Future versions may include ML models for risk prediction, disaster classification, or hazard forecasting. AI-powered heatmaps, pattern detection, and early warning forecasts can greatly improve users' situational awareness.
4. Advanced Data Visualisation (Heatmaps, Risk Zones, Evacuation Routes)
 - Enhanced visualization techniques, such as color-coded risk ratings, evacuation path mapping, and severity-based clustering, can help emergency responders better understand complex disaster data.

5. Offline-First Architecture.

- Expanding offline capabilities to include cached maps, previously received alerts, and locally saved hazard data would increase usability in remote or disaster-affected areas with limited connectivity.

6. User Reporting and Community Collaboration.

- Adding two-way communication tools allows users to report hazards, request assistance, and identify safe zones. Verified reports from the community can supplement official data sources and improve real-time accuracy.

7. Multilingual Support

- Adding Nepali and other regional languages can help make the app more accessible, particularly to communities with poor English ability.

Chapter 11: Conclusion

In summary, the goal of this project was to create a practical and impactful mobile application that would improve community preparedness and response during disasters. The Community Disaster Preparedness Application combines real-time notifications, geolocation services, disaster guidelines, and automated backend processing to give consumers timely and accurate information. The solution provides a seamless experience by combining Flutter for the frontend, Node.js, and Firebase for the backend. Global disaster data from GDACS is automatically fetched, processed, and pushed straight to users' smartphones. The program not only alerts users, but also provides them with critical preparedness tips that are accessible even when offline, assuring dependability during emergencies.

The initiative successfully demonstrates how new technologies may be used to improve public safety and raise awareness. Its clean interface, effective backend automation, and

dependable alerting system make it suitable for a wide range of users, independent of technical expertise. Although the program presently has limitations, such as Android-only support, reliance on internet access, and performance variations on low-end devices, it lays a good platform for future improvements. With planned enhancements such as iOS expansion, AI-based prediction, integration with local government APIs, improved visualization tools, and expanded offline functions, the system has enormous development potential.

Overall, the Community Disaster Preparedness Application is an important step toward accessible, technology-driven disaster management. It improves access to essential information, promotes preparedness, and leads to safer communities. By continuing to improve and expand its features, the program has the potential to become a comprehensive and indispensable tool for natural disaster awareness and emergency response.

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University Id: 2214169

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Chapter 13: Appendix

13.1 Weekly Reports of Undergraduate Project

UNIVERSITY OF BEDFORDSHIRE DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY UNDERGRADUATE PROJECT WEEKLY PROGRESS REPORT FORM	
Student's Name: <u>Aryam Bajracharya</u>	Supervisor's Name: <u>Pawan KC</u>
Project Title: <u>Community Disaster Preparedness</u>	
Week No: <u>3</u>	Report No: <u>1</u>
Summary of progress (including any problems)	- Setup the Flutter development environment. - Created basic UI pages (Login & Register) - Tested GIDACS API. - Faced issue with Flutter SDK path which was successfully resolved later.
Plan for next week	- Firebase for user reports & validation. - Create more pages (Alert, Profile, Report) - Design navigation for main pages (Home, Alert, Map, Checklist) - Integrate real-time alert data from GIDACS API.
Supervisor's comments	
Student's Signature <u>Aryam</u>	Supervisor's Signature <u>Pawan KC</u>
Date <u>10/10</u>	Date
When signed this form must be scanned and submitted via the relevant link on BREO	



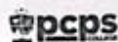
UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness	
Week No: 41	Report No: 2

Summary of progress (including any problems)	- Created firebase authentication for login & registration. - Designed home screen layout & navigated alert, maps, & emergency sections. - Started API integration with GIDACS to fetch live disaster & weather data.
Plan for next week	- Develop geolocation features using Google Maps API. - Display nearby shelters & emergency contacts using mock data. - Implement alert customization (selecting disaster types)
Supervisor's comments	

Student's Signature *Aryam* Supervisor's Signature *P.K.*
Date .. 47/10 Date

When signed this form must be scanned and submitted via the relevant link on BREO



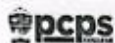
UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 5	Report No: 3

Summary of progress (including any problems)	- Implemented Cartodb Voyager Basemap from Carto. - Developed geolocation features. - Displayed emergency contacts & developed map. - Added real-time alerts using GDACS API.
Plan for next week	- Implement offline functionality using Flutter_map_tile_caching. - Create "Emergency Checklist" & "Profile" sections. - Begin incident reporting "Report".
Supervisor's comments	

Student's Signature *Aryam* Supervisor's Signature *Pawan*
Date *24/10* Date

When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 6	Report No: 4

Summary of progress (including any problems)	- Implement flutter library 'offline map tile caching'. - Developed safety methods for different disaster. - Created a single incident reporting & connected it to Firebase. - Testing offline map/mode.
Plan for next week	- Improve the incident reporting feature with image upload, disaster type, location, etc. - Add admin panel for verifying incident reports.
Supervisor's comments	

Student's Signature *Aryam* Supervisor's Signature *P.K.*
Date 31/10 Date

When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 7	Report No: 5

Summary of progress (including any problems)	- Enhanced the incident reporting feature (Critical report was also made for all users to see approved by admin) - Designed & implemented admin verification. - Tested & found minor errors when switching from online to offline mode.
Plan for next week	- Fixed the issues - Add OAuth 2.0 & data encryption for secure login. - Improve UI with good color palette & accessibility design.
Supervisor's comments	

Student's Signature *Aryam* Supervisor's Signature *Pawan*
Date *7/11* Date

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UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 8	Report No: 6

Summary of progress (including any problems)	- Resolved the issue & tested offline caching of map. - Integrated Google Authentication for secure login. - Received feedbacks from stakeholders (improved UI & alert layout)
Plan for next week	- Start testing documentation - Conduct more VAT (User Acceptance Testing) among friends for the app review. - Improve button, make it responsive, icons, etc.
Supervisor's comments	

Student's Signature *Aryam* Supervisor's Signature *Pawan*
Date .. 14 / 11 Date

When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 3	Report No: 7

Summary of progress (including any problems)	- Conducted app testing for users. - Improved button, icons, etc. - Managed ClickUp. - Checked notification while running the backend.
Plan for next week	- Prepare evaluation reports. - Work on fixing bugs & map errors.
Supervisor's comments	* Fix the bugs and work on the reports.

Student's Signature *Aryam* Supervisor's Signature *Pawan*
Date ... 22/11 Date

When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
UNDERGRADUATE PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Aryam Bajracharya	Supervisor's Name: Pawan KC
Project Title: Community Disaster Preparedness App	
Week No: 10	Report No: 8

Summary of progress (including any problems)	- Conducted final testing results & user feedback analysis. - Worked on app performance.
Plan for next week	- Prepare for presentation. - Make thesis, documentation (Final) - Submit the code, final report, etc.
Supervisor's comments	* Finalize the Report * Prepare the Presentation Material.

Student's Signature *Aryam* Supervisor's Signature *Pawan*
Date *28/11* Date *11/28/2028*

When signed this form must be scanned and submitted via the relevant link on BREO

13.2 Project Proposal

BSc Project Screening Form: Guidelines

Part 1 – Project Proposal

Student Name	Aryam Bajracharya
Student Number	2214169
Degree Pathway (course)	BSc (Hon)s CS & SE
Supervisor Name	Tara GC
Title of Project	Community Disaster Preparedness App
Abstract of the project	Nepal frequently suffers untimely natural disasters such as earthquakes, floods, and landslide which leads to several loss of life and property. Many people especially, in remote areas, they struggle to get timely notifications and access essential emergency resources due to poor internet connectivity and limited preparedness guidelines. This research paper will focus on disaster alert APIs, geolocation technologies to improve the efficiency of the proposed disaster preparedness application. Disaster alarm APIs from DHM Nepal, GDACS, and OpenWeather will be used to send accurate, real-time notifications. Navigation and geo-location services such as the Google Maps API and OpenStreetMap will help users locate shelters, emergency contacts and preparedness guide at the time of the disaster. The development process will involve integrating data from verified sources for real-time alerts. By implementing these research findings mobile app will be developed. This mobile application seeks to reduce risks, increase preparedness and support both urban and rural communities during emergencies by providing timely and reliable information. Keywords: Disaster alert systems, Risk reduction, Geolocation in disaster management, Offline emergency access, Emergency response, Mobile technology etc.
Project deliverables	- Functional Mobile Application - Poster - Contextual Report - Final Report/ Presentation
Description of your artefact	Problem Statement: "Many communities, especially in remote areas,

	people struggle in receiving timely alerts and access critical emergency due to lack of centralized information and poor connectivity which leads to delayed responses and increases risks during emergencies. Without proper warning and preparedness tools/ guides individuals are left vulnerable (UNDRR U. N., 2021)." ▪ Aim: - To enhance community disaster preparedness using technology and provide reliable and efficient disaster alert and response system. ▪ Objective: - Develop a mobile application with real-time alerts and notifications. - Make essential features available offline for reliability. - Provide emergency preparedness resources, including checklists, first aid guides and emergency contacts. - Implement secure data storage and authentication for user safety. ▪ List of features: - Real-time disaster alerts from verified sources. - Shelter locator with GPS and offline functionality. - Interactive preparedness checklists. - First aid and emergency response guide. - Contact details for emergency services including hospitals and rescue teams. - Secure login and user data management.			
Risk analysis	Risk Analysis API reliability issues affecting real-time alerts. High data dependency on	Impact High High	Likelihood Likely Likely	Mitigate risks Implement backup alert source for redundancy. (adding alternative source if primary alert system fails) Minimize external dependencies by

	external source. User engagement challenges. Cyber security threats and data privacy concerns. Accessibility issue in remote areas.	High High High	Likely Likely Highly Likely	storing local data. Develop interactive features to improve user engagement. Implement encryption, authentication protocols, and secure database management Ensure offline functionality and low-data mode.
How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired	During my BSc (Hons) in Computer Science and Software Engineering at PCPS College, I have learned important skills needed for building applications. I gained knowledge in Software Engineering Principles like backend development, API integration, and database management which helps to create efficient systems. I also learned developing Mobile Application using Flutter to build apps that run on both Android and iOS. Also, I studied Cloud Computing and Data Storage focusing on secure local storage and cloud systems to keep data safe and accessible. My coursework in UI/UX Designation taught me how to create simple, user friendly interfaces for better user experiences. These skills will help me develop my Community Disaster Preparedness App effectively, easy to access and useful in emergencies.			
Resources required in developing the artefact	- Development tools: Flutter, Firebase, SQL Lite, API services. - Cloud services: Databases for user data and alerts. - Testing devices: Mobile phones (Android/iOS). - Hardware: Personal Laptop with at least 8GB RAM and 100GB storage.			
Have you completed & submitted your ethics form?	YES ☒		NO	
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence				

Supervisor Signature:

27th Feb 2025

Aryam
2025/2/27

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section after your project proposal has been approved by your supervisor. Use Harvard referencing (see <https://lrweb.beds.ac.uk/a-guide-to-referencing>). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

- a. (W & Sena, 2013) *Disaster Prevention and Preparedness*
<https://www.humanitarianlibrary.org/sites/default/files/2013/07/DisasterPreventionPreparedness.pdf>
- b. (UNDRR, 2015) *Nepal Disaster Management reference handbook 2015*
<https://www.preventionweb.net/publication/nepal-disaster-management-reference-handbook-2015>
- c. (Cross, n.d.) *Community-Based Disaster Preparedness Training Manual*
<https://www.rcrc-resilience-southeastasia.org/wp-content/uploads/2017/07/1-DP-Manual-in-Eng.pdf>
- d. (UNDRR, PreventionWeb, 2019) *Disaster Risk Reduction in Nepal*
https://www.preventionweb.net/files/68257_682306nepaldrmstatusreport.pdf

2. Journal Papers

- a. (Galasso, 2019) *International Journal of Disaster Risk Reduction*
<https://www.sciencedirect.com/journal/international-journal-of-disaster-risk-reduction>
- b. (Belgrade, 2024) *International Journal of Disaster Risk Management*
<https://internationaljournalofdisasterriskmanagement.com/index.php/Vol1>
- c. (Alexander, 2016) *Emergency Preparedness and Response*
https://www.researchgate.net/publication/371108859_Acceptance_of_Mobile_Application_on_Disaster_Preparedness_Towards_Decision_Intelligence_in_Disaster_Management

- d. (Syukron, Madugalla, Mojtaba , & Grundy, 2024)A Comprehensive Study of Disaster Support of Mobile Apps
https://www.researchgate.net/publication/382178486_A_Comprehensive_Study_of_Disaster_Support_Mobile_Apps

3. Web Sites with relevant information

- a. https://eiscouncil.org/ceo/qinom-disaster-preparedness-simulations/?utm_source=google_ads&utm_medium=22055380472&ad_group=&ad=&device=c&keyword=&qad_source=1&qclid=CjwKC_AiA2JG9BhAuEiwAH_zf3pMylL-pzZjiiK6PtyE0u2UOKeBf8FqhF_IVheGD22mMJNEaeMmX7hoCVeQQAvD_BwE
- b. <https://www.ready.gov/be-informed>
- c. <https://www.preventionweb.net/publication/nepal-disaster-management-reference-handbook-2015>
- d. <https://www.ready.gov/be-informed>

4. Relevant software

- a. Flutter/React Native
- b. Visual Studio Code
- c. Visual Paradigm Community Edition

5. Relevant hardware

- a. Personal Laptop: A device that supports sufficient processing power and memory to run machine learning algorithms and development tools (8 GB RAM or more, 1920 x 1080 minimum screen resolution).
- b. 100 GB of available disk space minimum (Windows Installation, IDE + Android SDK + Android Emulator, and other necessary software).

6. Other

- a. ...
- b. ...

13.3 Ethics Form

FACULTY OF CREATIVE ARTS, TECHNOLOGIES AND SCIENCE

Form for Research Ethics Projects (Ethics Form)

Student Name	Aryam Bajracharya
Student Number	2214169
Degree Pathway	BSc(Hon)s CS & SE
Supervisor name	Tara GC
Supervisor Signature	
Title of project	Community Disaster Preparedness App

SECTION A

This project focuses on developing a Community Disaster Preparedness App that provides real-time alerts, emergency contacts and offline access to help communities during disasters. In this research there are no major ethical issues related to Community Disaster Preparedness App. All the collected data will be kept confidential with no personal details shared. Data will be stored safely to prevent unauthorized access.

SECTION B Check List

Please answer the following questions by circling YES or NO as appropriate.

Does the study involve vulnerable participants or those unable to give informed consent (e.g. children, people with learning disabilities, your own students)?	YES
	NO ✓
Will the study require permission of a gatekeeper for access to participants (e.g. schools, self-help groups, residential homes)?	YES
	NO ✓
Will it be necessary for participants to be involved without consent (e.g. covert observation in non-public places)?	YES
	NO ✓
Will the study involve sensitive topics (e.g. obtaining information about sexual activity, substance abuse)?	YES
	NO ✓
Will blood, tissue samples or any other substances be taken from participants?	YES
	NO ✓
Will the research involve intrusive interventions (e.g. the administration of drugs, hypnosis, physical exercise)?	YES
	NO ✓
Will financial or other inducements be offered to participants (except reasonable expenses or small tokens of appreciation)?	YES
	NO ✓
Will the research investigate any aspect of illegal activity (e.g. drugs, crime, underage alcohol consumption or sexual activity)?	YES
	NO ✓
Will participants be stressed beyond what is considered normal for them?	YES
	NO ✓
Will the study involve participants from the NHS (patients or staff) or will data be obtained from NHS premises?	YES
	NO ✓

If the answer to any of the questions above is "Yes", or if there are any other significant ethical issues, then further ethical consideration is required. Please document carefully how these issues will be addressed.

Signed (student): Aryam

Date: February 27, 2025

Countersigned (Supervisor):

Date:

27th Feb, 2025

13.4 Google Form Questionnaire

The image shows a Google Form titled "Community Disaster Preparedness App". At the top, there are tabs for "Questions", "Responses" (with a count of 170), and "Settings". The form is divided into sections, with the first section labeled "Section 1 of 4". The title "Community Disaster Preparedness App" is displayed in a large font, followed by a description: "This mobile application helps people to be disaster ready. It sends real-time alerts from verified sources, enables offline access to emergency resource such as shelter locations, safety checklists and helps users to find nearby emergency services." Below the description, there are two questions, both marked as required with a red asterisk. The first question is "What is your age group?" with four radio button options: "Below 18", "18 - 30", "31 - 50", and "Above 50". The second question is "What is your gender?" with four radio button options: "Male", "Female", "Prefer not to say", and "Other:" followed by a text input field. On the right side of the form, there is a vertical toolbar with icons for adding questions, duplicating, deleting, and other editing functions.

Questions Responses 170 Settings

Section 1 of 4

Community Disaster Preparedness App

This mobile application helps people to be disaster ready. It sends real-time alerts from verified sources, enables offline access to emergency resource such as shelter locations, safety checklists and helps users to find nearby emergency services.

What is your age group? *

☐ Below 18

☐ 18 - 30

☐ 31 - 50

☐ Above 50

What is your gender? *

☐ Male

☐ Female

☐ Prefer not to say

☐ Other: _____

Which type of area do you live in? *

☐ Urban

☐ Rural

Have you experienced a natural disaster (e.g., earthquake, flood, landslide)? *

☐ Yes

☐ No

After section 1 Continue to next section

Section 2 of 4

Disaster Preparedness Awareness

Description (optional)

Are you aware of any mobile apps for disaster preparedness? *

☐ Yes

☐ No

Do you currently receive any disaster alerts on your phone (via SMS or apps)? *

☐ Yes

☐ No

Which of the following disaster features do you think are most important? (Select those you think that are important) *

- ☐ Real-time alerts
- ☐ Offline accessibility
- ☐ Shelter locator
- ☐ Emergency checklists
- ☐ First aid guides

How frequently do you check for disaster alerts or updates? *

- ☐ Frequently
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

Section 3 of 4

Mobile App Usability & Features



Description (optional)

How often do you use mobile apps to get disaster information? *

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Rarely

Would you prefer an app that works offline in case of network failure? *

- ☐ Yes
- ☐ No

How important is it for the app to include geolocation features (finding nearby shelters)? *

- ☐ Very Important
- ☐ Somewhat Important
- ☐ Not Important

Do you find personalized safety checklists useful during emergencies? *

☐ Yes

☐ No

After section 3 Continue to next section ▼

Section 4 of 4

Feedback & Suggestions ▼ ⋮

Description (optional)

What additional features would you like to see in a disaster preparedness app?

Long-answer text

What challenges have you faced in receiving timely disaster alerts or emergency information?

Long-answer text

Would you be interested in testing this app when it becomes available?

☐ Yes

☐ No

Do you have any concerns about using mobile technology for receiving disaster alerts?

Long-answer text

Any additional comments or suggestion?

Short-answer text

Form_Responses1						
Timestamp	What is your age group?	What is your gender?	Which type of area do you live in?	Have you experienced a natural disaster (e.g	Are you aware of any mobile apps for disaster	Do you currently receive any disaster alerts?
08/03/2025 12:16:00						
08/03/2025 12:19:10	18 - 30	Male	Rural	Yes	No	Yes
08/03/2025 12:34:08	18 - 30	Female	Urban	Yes	No	No
08/03/2025 12:51:39	18 - 30	Male	Urban	Yes	Yes	No
08/03/2025 12:55:16	18 - 30	Female	Urban	Yes	Yes	Yes
08/03/2025 12:57:11	18 - 30	Female	Urban	Yes	Yes	No
08/03/2025 12:59:36	18 - 30	Male	Urban	Yes	No	No
08/03/2025 12:59:53	18 - 30	Male	Urban	Yes	No	No
08/03/2025 13:01:31	18 - 30	Male	Rural	Yes	No	No
08/03/2025 13:01:34	18 - 30	Male	Urban	Yes	Yes	Yes
08/03/2025 13:02:05	18 - 30	Male	Urban	Yes	No	No
08/03/2025 13:05:09	18 - 30	Female	Urban	Yes	No	Yes
08/03/2025 13:06:38	31 - 50	Female	Urban	Yes	No	No
08/03/2025 13:07:06	Below 18	Male	Urban	Yes	No	No
08/03/2025 13:08:17	18 - 30	Male	Urban	Yes	No	Yes
08/03/2025 13:15:02	Below 18	Male	Urban	Yes	No	Yes
08/03/2025 13:20:38	18 - 30	Male	Urban	Yes	Yes	No
08/03/2025 13:36:51	18 - 30	Male	Urban	Yes	Yes	No
08/03/2025 13:37:03	18 - 30	Female	Urban	Yes	Yes	Yes
08/03/2025 13:43:11	18 - 30	Male	Urban	Yes	No	No
08/03/2025 13:46:49	18 - 30	Male	Rural	Yes	No	No
08/03/2025 14:43:43	31 - 50	Male	Urban	Yes	No	No
08/03/2025 14:51:22	18 - 30	Male	Urban	Yes	Yes	Yes
08/03/2025 15:00:01	18 - 30	Male	Urban	Yes	No	No
08/03/2025 15:22:50	18 - 30	Female	Urban	Yes	Yes	No
08/03/2025 16:28:11	18 - 30	Female	Urban	Yes	Yes	No

Student Name: Aryam Bajracharya Community Disaster Preparedness App
University Id: 2214169

Which of the following disaster features do you want?	How frequently do you check for disaster alerts?	How often do you use mobile apps to get disaster alerts?	Would you prefer an app that works offline or online?	How important is it for the app to include geolocation features?	Do you find personalized safety checklists useful?
Real-time alerts, Offline accessibility, Shelter location					
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Somewhat Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Somewhat Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Monthly	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Monthly	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Weekly	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Never	Rarely	Yes	Very Important	No
Real-time alerts, Offline accessibility, Shelter location	Never	Daily	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Rarely	Yes	Somewhat Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Rarely	Yes	Very Important	No
Real-time alerts, Offline accessibility, Shelter location	Never	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Daily	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	No
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Somewhat Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Occasionally	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Frequently	Daily	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Frequently	Daily	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Very Important	Yes
Real-time alerts, Offline accessibility, Shelter location	Rarely	Rarely	Yes	Somewhat Important	Yes

What additional features would you like to see?	What challenges have you faced in receiving disaster alerts?	Would you be interested in testing this app?	Do you have any concerns about using mobile apps for disaster alerts?	Any additional comments or suggestions?	How frequently do you check for disaster alerts?
Anything	Not accurate	Yes	Offline mode and timely accurate alerts	Good luck	
Available nearby hospitals or clinics, available geolocation	N/A	Yes	N/A	N/A	
Real-time chat	Delay information	Yes	Yes	No	
Family members location	Network issues	Yes	Should be from trusted source	No	
No	Stayed out of house	Yes	Ok	Great	
Alert Alarm	Never got any	Yes	No	No	
Alert	Safety	Yes	No	No	
Facebook	Nothing	Yes	No	No	
Perfect location	User should get notification before disaster	Yes	It should be user friendly	Protect equipment and materials	
I've never used one before so I need to do some training	Never got one.	Yes	No I think it's a great idea	I need to use the app first.	
direct contact features to authorities, disaster management	late arrival of relief/ help	Yes	network traffic might be an issue	nope	
I would like to see some shelters around my area	I panic when there is a disaster and get really scared	Yes	I do not have any concerns	The alert needs to come at least 10 minutes before	
Subscription less	Idk	Yes	Yes	Nope	
Preventive measures that anyone can use anywhere	Been in the middle of the storm. Got notification late	Yes	N/A	Geophysical location should be precise. The first	
Information	Network problem, Signal problem	Yes	No	Nice idea	
Accuracy	Inaccurate information	Yes	Not at all	Great work	
Predicament	Delay in information, no notification	Yes	No	No	
Alert, shelter nearby in emergency	Network issues	Yes	Yes	No	
One-Tap Safety Status	Lack of Real-Time Updates	Yes	Yes	Better Offline Support	
Emergency alerts	Nothing	Yes	No	Nothing	
Be able to contact people using any available channels	Haven't received any information	Yes	No	When will it be available	
N/a	N/a	No	N/a	N/a	
personalized emergency plans, real-time alerts, of network congestion during emergencies, and lack of		Yes	Yes, there are a few concerns about relying solely on mobile apps. Not everyone owns a smartphone or has it charged. Power outages during disasters can make it difficult to use mobile apps. It's important to have multiple alert systems in place.	To improve disaster alert systems, Public education is key.	
Offline service	Disaster alerts	Yes	Yes	No	
Alert	Nothing	Yes	No	N	

Google Form Link

<https://docs.google.com/forms/d/e/1FAIpQLSdPgs800smE9b36W2k7vNysyMYUsQqKQARm4cXLKySFEy-9mw/viewform?usp=dialog>

Student Name: Aryam Bajracharya Community Disaster Preparedness App
University Id: 2214169

Google Form Response Link

https://docs.google.com/spreadsheets/d/1q_OD_h5m4xbVwnPIOHCONPH9_nF24nfUfLDnq9WV4-w/edit?resourcekey=&gid=1420403851#gid=1420403851

Figma Link

<https://www.figma.com/design/cKVY3ORARJFt8y76OjiTXI/Untitled?node-id=0-1&t=sNpgLloCSPWjqIvk-1>